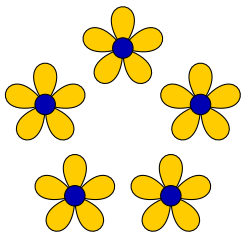
# Middle Primary Division

## Questions 1 to 10 are worth 3 marks each.

**1.** What is the total number of petals on all 5 flowers?



(A) 10 (B) 15 (C) 20 (D) 25 (E) 50

$5 \times 5 = 25$,

hence (D).

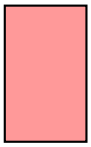
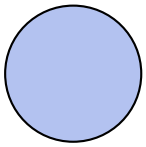
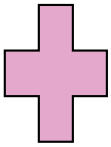
**2.** $2 + 3 + 7 + 8 =$

(A) 10 (B) 20 (C) 30 (D) 40 (E) 50

We can pair $2 + 8 = 10$ and $3 + 7 = 10$ so that $2 + 3 + 7 + 8 = 20$,

hence (B).

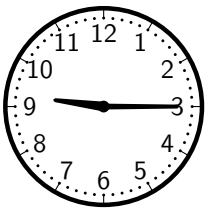
**3.** Which one of these shapes is a rectangle?

(A)  (B) (C) (D)  (E) 

Only (A) is a rectangle,

hence (A).

**4.** Which digital clock time matches the time shown on the clock face?




(A) 3:09 (B) 9:03 (C) 9:15 (D) 10:03 (E) 10:15

The time is 15 minutes past 9, or 9:15,

hence (C).

**5.** Emma has a bag containing 5 red, 4 yellow, 1 black and 2 blue buttons. When she chooses 1 button at random, what colour is it most likely to be?

(A) green (B) blue (C) black (D) yellow (E) red

The most common button colour is red, so this is most likely,

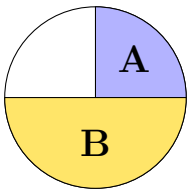
hence (E).

---

**6.** What fraction of the circle is part A?

(A) one-half (B) one-third (C) two-thirds

(D) one-quarter (E) three-quarters



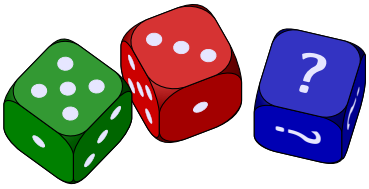

Split B into two equal parts, then the circle is divided into 4 equal quarters, and A is one of these quarters,

hence (D).

---

**7.** In a board game, Nik rolls three standard dice, one at a time. He needs his three rolls to add to 12.

His first two dice rolls are 5 and 3. What does he need his third roll to be?



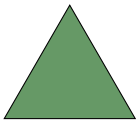
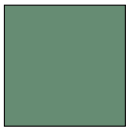
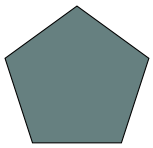
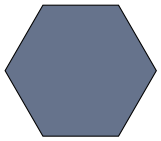
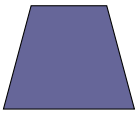

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

$5 + 3 = 8$ and $12 - 8 = 4$,

hence (C).

---

**8.** Which shape has *only one* line of symmetry?

(A)  (B)  (C)  (D)  (E) 

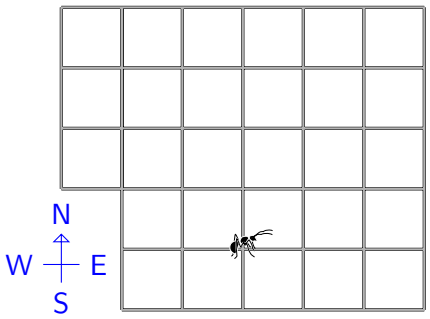
Choices (A) to (D) are all regular polygons, so have a line of symmetry through each vertex. However (E) only has one line of symmetry,

hence (E).

---

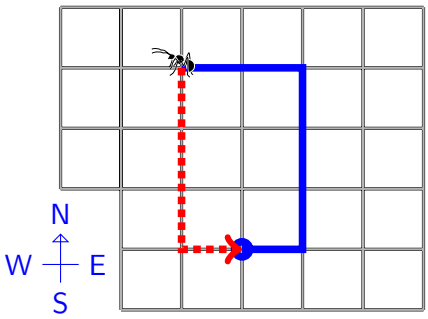
**9.** Anastasia Ant is on a sheet of wire mesh with 10 cm squares. She can only move along the wires. She moves 10 cm east, then 30 cm north and 20 cm west.

What is the least possible distance she needs to move to return to her starting point?




(A) 10 cm (B) 20 cm (C) 30 cm (D) 40 cm (E) 50 cm

Her outgoing journey is shown in solid blue:



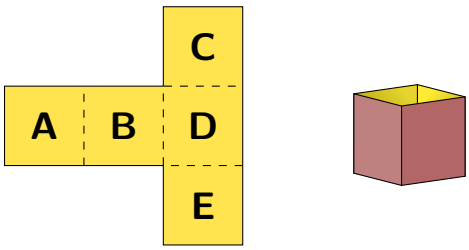

Then she is 30 cm north and 10 cm west of her starting position. So she must travel at least 40 cm to return to the start, as the dotted red path shows,

hence (D).

**10.** A piece of card is cut out and labelled as shown in the diagram.

It is folded along the dotted lines to make a box without a top.

Which letter is on the bottom of the box?




(A) A (B) B (C) C (D) D (E) E

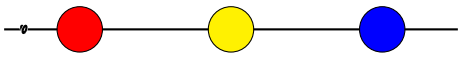
Face B is adjacent to A and D in the net, and becomes adjacent to C and E once folded. Since the only face adjacent to all 4 other faces is the bottom of the box, this bottom face must be labelled B,

hence (B).

## Questions 11 to 20 are worth 4 marks each.

**11.** Jiang is threading beads onto a string to make a necklace. He decides to use a sequence, starting with a red bead, then a yellow bead and a blue bead. He repeats this pattern until he has 20 beads on the necklace.

How many red beads will he use?



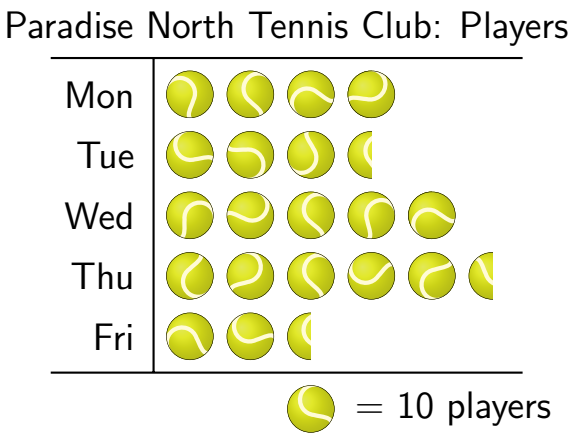
(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

Number the beads $1, 2, 3, 4, \ldots$, then beads 1, 4, 7, 10, 13, 16 and 19 are red, making 7 red beads,

hence (C).

**12.** How many people played tennis on the busiest day of the week at the Paradise North Tennis Club?

(A) 45 (B) 50 (C) 55 (D) 60 (E) 65

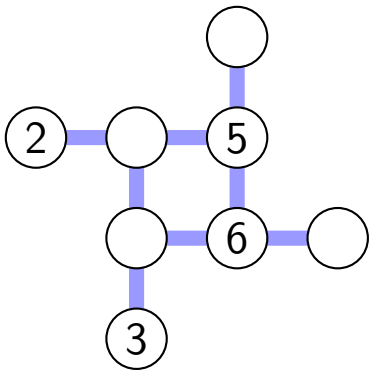



The busiest day was Thursday. Each ball in the chart represents 10 players. So the 5 whole balls represent 50 and the extra half ball represents 5 more, giving 55 players in total,

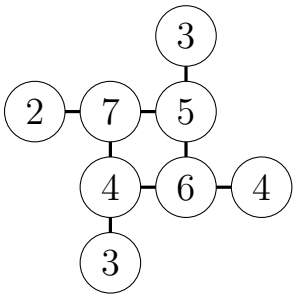
hence (C).

**13.** When this puzzle is complete, each of the four lines will add up to 14. What will the four missing numbers add to?

(A) 10 (B) 14 (C) 15 (D) 16 (E) 18




(AK) In the right-hand vertical column, $3 + 5 + 6 = 14$.
In the higher horizontal row, $2 + 7 + 5 = 14$.
In the left-hand vertical column $7 + 4 + 3 = 14$.
In the lower horizontal row, $4 + 6 + 4 = 14$.




So the four missing numbers were 3, 7, 4 and 4 and their sum is $3 + 7 + 4 + 4 = 18$,

hence (E).

---

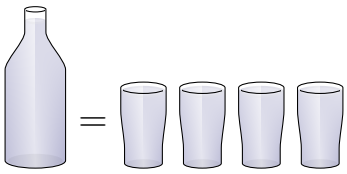
**14.** Which one of the following could give an answer of 25?

(A) an odd number plus an odd number
(B) an even number multiplied by an even number
(C) an even number divided by an odd number
(D) an odd number minus an even number
(E) an even number multiplied by an odd number

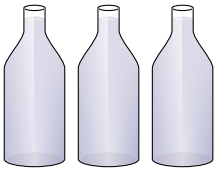
It is possible for an odd number minus an even number to be 25, such as $35 - 10 = 25$.
All other choices in the question are guaranteed to be even,

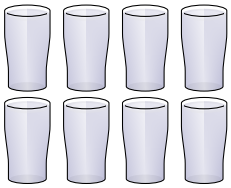
hence (D).

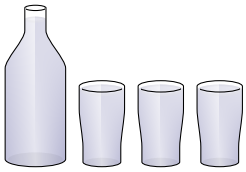
**15.** This bottle holds 4 glasses of water.

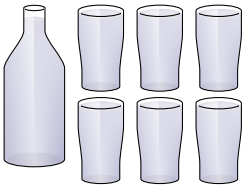


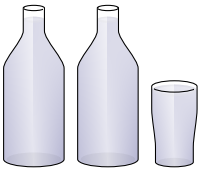
Which one of the following holds the most water?

(A) 

(B) 

(C) 

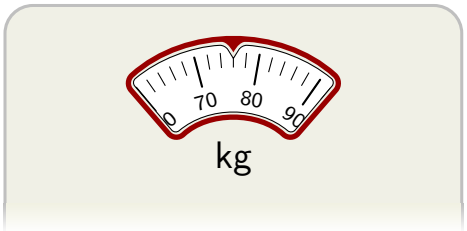
(D) 

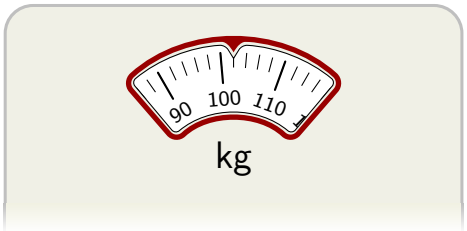
(E) 

---

Measured in glasses of water, (A) holds $3 \times 4 = 12$ glasses, (B) holds 10 glasses, (C) holds $4 + 3 = 7$ glasses, (D) holds $4 + 6 = 10$ glasses and (E) holds $4 + 4 + 1 = 9$ glasses. So (A) holds the most water, hence (A).

**16.** Fido needs to be weighed at the vet, but he won't sit still. So the vet weighs himself and then weighs himself holding Fido.




How much does Fido weigh?

(A) 23 kg (B) 26 kg (C) 28 kg (D) 30 kg (E) 32 kg

The minor tickmarks separate each block of ten kilograms into five equal intervals of 2 kg each. By himself, the vet weighs 76 kg. When holding Fido, he weighs 102 kg. So Fido weighs $102 - 76 = 26$ kg,

hence (B).

---

**17.** Grandma adds 3 spoons of sugar to the juice of 10 lemons to make lemonade. How many lemons are needed if 15 spoons of sugar are used to make Grandma's recipe?

(A) 20 (B) 22 (C) 25 (D) 30 (E) 50

The recipe has 3 spoons of sugar to 10 lemons. The batch with 15 spoons of sugar has 5 times as much sugar, so it will have 5 times as many lemons, which is $5 \times 10 = 50$ lemons,

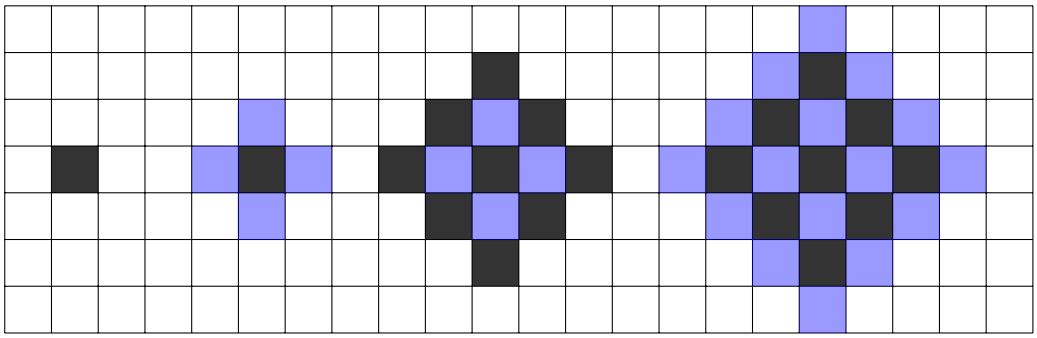
hence (E).

---

**18.** Janus is making patterns using square tiles.

Each pattern is made by copying the previous pattern, then adding new tiles around the outside edges.

The new tiles alternate in colours, as shown.



Pattern 1 Pattern 2 Pattern 3 Pattern 4

What will be the total number of tiles in Pattern 5?

(A) 25 (B) 33 (C) 37 (D) 40 (E) 41

---

Pattern 5 has the existing 16 blue and 9 grey tiles, and then 16 additional grey tiles are added around the edge. So pattern 5 has $16 + 9 + 16 = 41$ tiles,

hence (E).

---

**19.** My laptop's battery meter tells me what fraction of the battery charge is left. I used the laptop for 4 hours and 20 minutes and it went from

to

What would the reading be after using the laptop for another 130 minutes?

(A)

(B)

(C)

(D)

(E)

---

4 hours and 20 minutes is 260 minutes, during which time 4 lights went off. So in the next 130 minutes, another 2 lights will go off,

hence (B).

---

**20.** Syed's mother had some money to share with her family.
She gave one-quarter of her money to Syed.
Then she gave one-third of what was left to Ahmed.
Then she gave one-half of what was left to Raiyan.
She was left with \$15, which she kept for herself.
How much money did Syed's mother have to start with?

(A) \$30 (B) \$45 (C) \$60 (D) \$90 (E) \$120

*Alternative 1*

Working backwards, the \$15 she at the end was after she gave half of what she had to Raiyan. So she had \$30 before giving to Raiyan.

This \$30 was after giving one-third to Ahmed, so \$30 is two-thirds of what she had before that, which means she must have given Ahmed \$15 out of the \$45 she had before giving to Ahmed.

This \$45 is after giving one-quarter of her original money to Syed. So \$45 was three-quarters of her original money, meaning that one-quarter was \$15 and her original amount was \$60,

hence (C).

*Alternative 2*

For every \$1 that she started with, she first gives 25 cents to Syed, leaving her with 75 cents. Then 25 cents to Ahmed, leaving her with 50 cents. Then 25 cents to Raiyan, leaving her with 25 cents. But she was left with \$15, with is 60 times as much as this. So she must have started with \$60 and each family member received \$15,

hence (C).

## Questions 21 to 25 are worth 5 marks each.

**21.** In 2023, Janine's birthday is on a Thursday. Ngoc's birthday is one month later, meaning that it is on the same day-number in the next month. Ngoc's birthday is not on a Saturday or a Sunday. Which day of the week is Ngoc's birthday?

(A) Monday (B) Tuesday (C) Wednesday (D) Thursday (E) Friday

Since 2023 is not a leap year, the month of Janine's birthday has 28, 30 or 31 days. 28 days after Janine's birthday is exactly 4 weeks, so a Thursday. Counting on from this, 30 days after Janine's birthday is Saturday and 31 days is Sunday, which are both ruled out. So the only possibility is that Ngoc's birthday in 2023 is a Thursday,

hence (D).

**22.** Sally empties her piggy bank to see how much she has saved. She has 20 coins altogether and the total value of her coins is \$3.80. She notices that she has twice as many 50c coins as 20c coins. The remaining coins are 10c coins. How many 10c coins does she have?

(A) 2 (B) 4 (C) 8 (D) 11 (E) 14

$2 \times 50 + 1 \times 20 + 17 \times 10 = 290$

$4 \times 50 + 2 \times 20 + 14 \times 10 = 380$

any more larger coins gives a larger total,

hence (E).

**23.** Three different numbers from 1 to 20 are chosen. They add to 20. Two of the numbers are 2 apart. How many possibilities are there for the largest number?

(A) 6 (B) 7 (C) 8 (D) 9 (E) 10

---

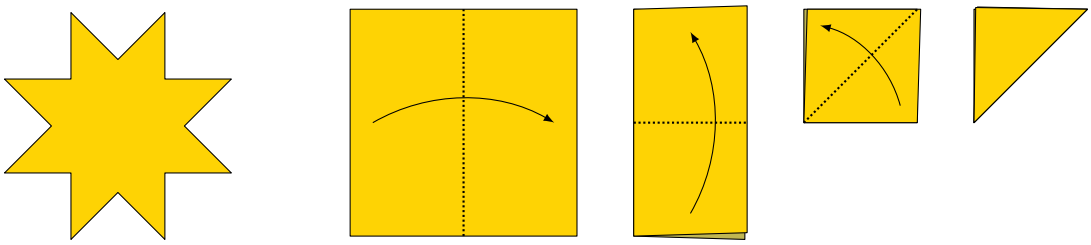
Consider choosing the numbers that are two apart first, starting from $1+3$, then $2+4$ and so on.

$1+3+16 = 2+4+14 = \cdots = 5+7+8$ gives 5 possibilities: the largest number could be 16, 14, 12, 10 or 8.

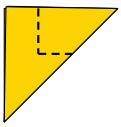
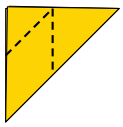
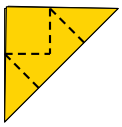
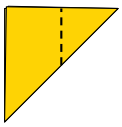
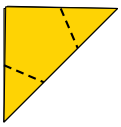
Then $6+8+6 = 20$ repeats 8, $7+9+4 = 20$ shows that the 6th possibility is that the largest number is 9. and finally $8+10+2 = 20$ repeats 10. Beyond this, the first two numbers already add to 20 or more, so there are no more possibilities,

hence (A).

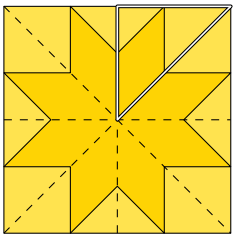
**24.** Estelle is making decorations shaped like the 8-pointed star shown. She folds a square of paper to make a triangle with 8 layers as shown.



How could she cut the triangle so that the unfolded shape is the star?

(A)  (B)  (C)  (D)  (E) 

Draw the page as if it had been unfolded without cutting, with the star on top:



In the outlined triangle, the paper that is part of the star is a rhombus, matching choice (B),

hence (B).

**25.** Tom is digging holes in his garden for his new plants. As he is working he slows down. Each new hole takes a minute longer to dig than the last hole.

The first 5 holes have taken him 35 minutes in total. How long will it take him to dig the next 2 holes?

(A) 11 minutes (B) 17 minutes (C) 21 minutes (D) 24 minutes (E) 25 minutes

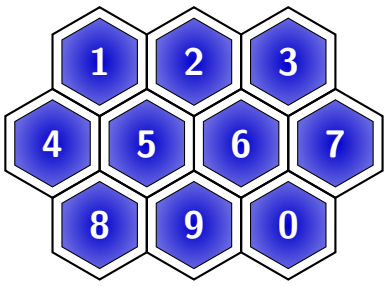
The first five holes are dug in an average of 7 minutes each. Since they increase by 1 minute per hole, they take $5, 6, 7, 8$ and 9 minutes. The next two holes take $10 + 11 = 21$ minutes,

hence (C).

**For questions 26 to 30, colour in the bubbles on the answer sheet to record whole-number answers from 0 to 999.**
**See the back cover for more details.**
**Questions 26 to 30 are worth**
**6, 7, 8, 9 and 10 marks respectively.**

---

**26.** To unlock a security door, Daniyal enters a code on the keypad shown. To make it easy to remember:




- the code has 10 digits
- the code uses every digit exactly once
- the code steps from button to button only if the two buttons touch
- the code is as large as possible.

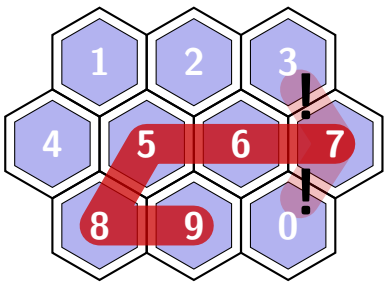
What are the last 3 digits in Daniyal's code?

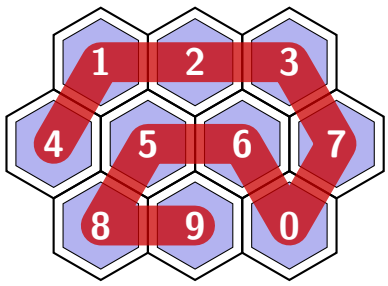
---

*Alternative 1*

To make the largest number we start with digit 9. We then continue to the next highest adjacent button for as long as this doesn't cut off any other buttons on the keypad. So the first four digits of the code are 9856.

If we try to go to 7 next, this will cut off the 0 or the other four buttons, as shown on the left, and the code cannot be completed. Similar issues arise if we try to go to 3 or 2 next.




So the next button must be 0 and the rest of the code can be completed as shown on the right,

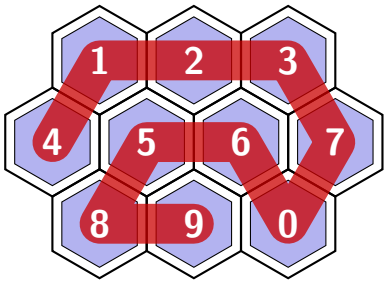
hence (214).

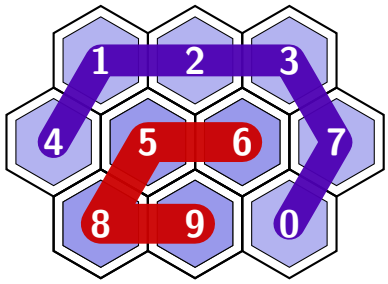
*Alternative 2*

(CW/AK) By starting at 9 and always choosing the largest neighbouring button, the path 985673214 can be found. However this path misses button

0, so the modified path 9856073214, shown below left, seems likely to be the best (largest) solution.

To confirm that 9856073214 is best, notice that no solution can have a better first 4 digits than 9856. Once we have used 9856, the rest of the path must string together the remaining six buttons, either as 073214 (possible) or 412370 (impossible).



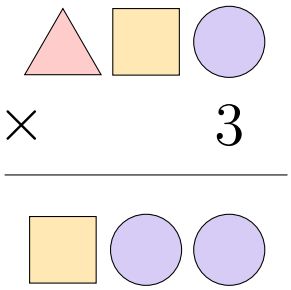




The only complete path is then 9856073214, which must be the best, hence (214).

---

**27.** In this puzzle, △, □ and ○ represent different nonzero digits.

What is the three-digit number △□○?




There must be ○ = 5 in the units column. Then there is a carry of 1 from the units to the tens column.

In the tens column the last digit of □ × 3 must be 4. That is, □ = 8. Then there is a carry of 2 from the tens to the hundreds column.

In the hundreds column △×3 = 6 so that △ = 2. In all, △□○ = 285, hence (285).

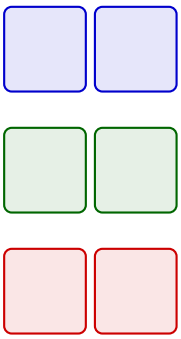
---

**28.** Ollie arranges the numbers 1 to 6 to form three 2-digit numbers.

Exactly one of his numbers divides by 3 and exactly one other divides by 4. None of them divide by 5.

The three numbers add to 93.

What is the number which does not divide by either 3 or 4?



(AK) The 5 must be in the tens place of one of the numbers.

To stay less than 110, the 6 must be in the ones place of one of the numbers.

The total of the ones places must be 13, which can only be done as $3 + 4 + 6 = 13$, and then the tens places must be $1 + 2 + 5 = 8$.

To get a multiple of 4, either 16, 24 or 56. However, if 24 is one of the other numbers, then the 6 must be in either 16 or 56, so 24 is not one of the numbers.

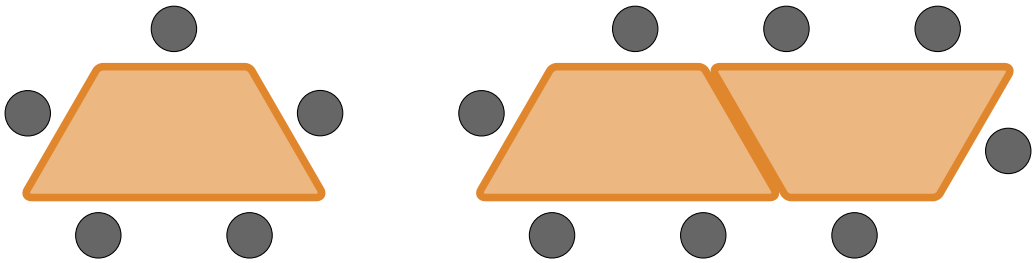
To get a multiple of 3, either 24 or 54. But as noted, 24 can't be one of the numbers, so 54 is the multiple of 3.

Then the multiple of 4 must be 16

The number that is not divisible by 3 nor 4 is 23,

hence (23).

**29.** A table in the shape of a trapezium can seat 5 people. When two tables are put together in a row, 8 people can be seated. What is the smallest number of trapezium tables required to seat 2023 people if they are all placed in a row?



(AK) After the first table of 5, each new table increases the number of seats by 3. So we look to solve $5 + 3 + 3 + \cdots = 2023$.

Then the number of tables past the first one is at least $(2023 - 5) \div 3 = 2018 \div 3 = 672\frac{2}{3}$. So it can be done with 673 tables after the first one, and then the minimum number of tables is 674,

hence (674).

**30.** Yifan has a construction set consisting of red, blue and yellow rods. All rods of the same colour are the same length, but differently coloured rods are different lengths. She wants to make quadrilaterals using these rods.

- When she uses two red, one blue and one yellow rod, the perimeter of the quadrilateral is 36 cm.
- When she uses two blue, one red and one yellow rod, the perimeter is 35 cm.
- When she uses two yellow, one blue and one red rod, the perimeter is 33 cm.

What number do you get when you multiply the lengths of one red rod, one blue rod and one yellow rod?

---

(AK) Label the rods R, B and Y.

By comparing the first and second quadrilaterals, R is 1 cm longer than B

By comparing the first and third quadrilaterals, R is 3cm longer than Y

So a square with 4 R's would have perimeter $36 + 1 + 3 = 40$. Consequently $R = 10$, and then $B = 9$ and $Y = 7$.

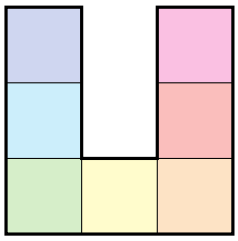
Multiplying $R$, $B$ and $Y$ gives $10 \times 9 \times 7 = 630$,

hence (630).

# Upper Primary Division

## Questions 1 to 10 are worth 3 marks each.

**1.** This shape is made from 7 squares, each 1 cm by 1 cm. What is its perimeter?



(A) 7 cm (B) 12 cm (C) 14 cm (D) 16 cm (E) 28 cm

*Alternative 1*

Counting from top left and going clockwise, the perimeter is $1+2+1+2+1+3+3+3 = 16$ cm,

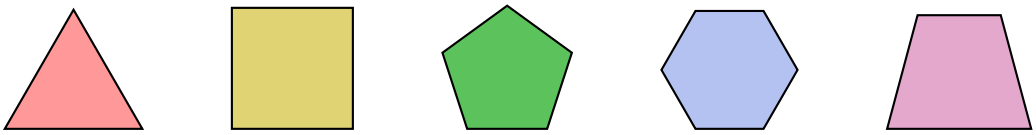
hence (D).

*Alternative 2*

Starting with one square of perimeter 4 cm each additional square removes 1 *cm* of perimeter and adds 3 cm, an increase of 2 cm. After adding 6 additional squares to the first one, the perimeter is $4+6\times 2 = 16$ cm,

hence (D).

**2.** There are five shapes here. How many are quadrilaterals?

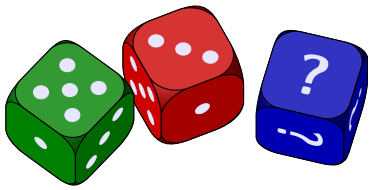


(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

The second (yellow square) and last (purple trapezium) are the only quadrilaterals,

hence (B).

**3.** In a board game, Nik rolls three standard dice, one at a time. He needs his three rolls to add to 12.



His first two dice rolls are 5 and 3. What does he need his third roll to be?

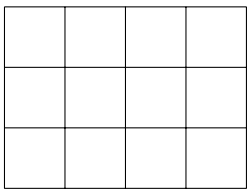
(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

$5 + 3 = 8$ and $12 - 8 = 4$,

hence (C).

---

**4.** In this diagram, how many of the small squares need to be shaded for the large rectangle to be one-quarter shaded?



(A) 2 (B) 3 (C) 4 (D) 6 (E) 12

There are 12 small squares, and one quarter of this is 3 small squares,

hence (B).

---

**5.** Petra left for school at 8:51 am. She got to school at 9:09 am.

How long did it take Petra to get to school?

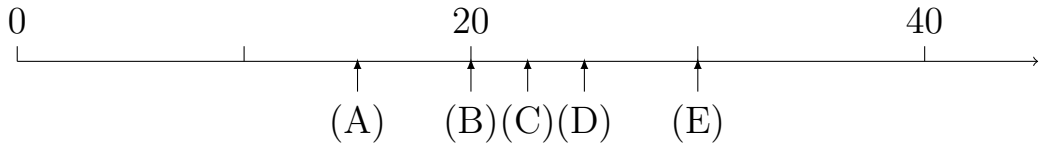
(A) 9 minutes (B) 10 minutes (C) 18 minutes (D) 42 minutes (E) 1 hour

Her trip included the 9 minutes before 9am then the 9 minutes after 9am, making a total of 18 minutes,

hence (C).

---

**6.** Which letter marks where 25 is on this number line?

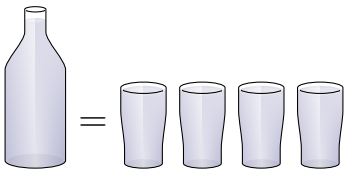



Point (E) marks the point half-way between 20 and 40, which is at 30. Then Point (D) marks the point half-way between 20 and 30, which is at 25,

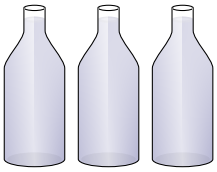
hence (D).

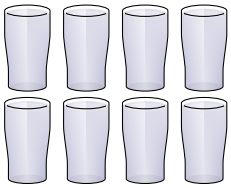
---

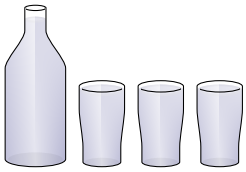
**7.** This bottle holds 4 glasses of water.

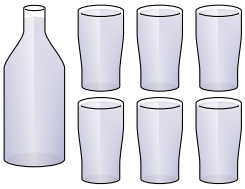


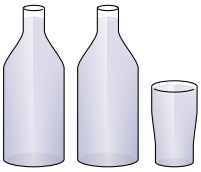
Which one of the following holds the most water?

(A) 

(B) 

(C) 

(D) 

(E) 

Measured in glasses of water, (A) holds $3 \times 4 = 12$ glasses, (B) holds 10 glasses, (C) holds $4 + 3 = 7$ glasses, (D) holds $4 + 6 = 10$ glasses and (E) holds $4 + 4 + 1 = 9$ glasses. So (A) holds the most water,

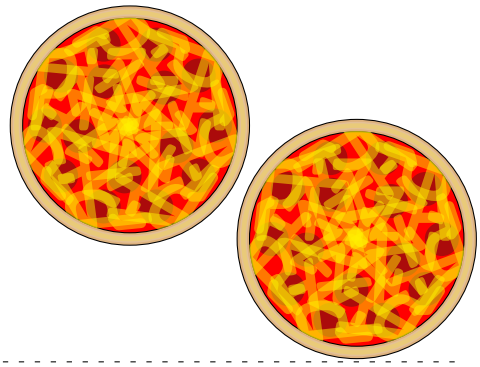
hence (A).

---

**8.** Two pizzas are shared equally between 3 students.

What fraction of a whole pizza does each student get?

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) $\frac{2}{3}$ (E) $\frac{3}{4}$



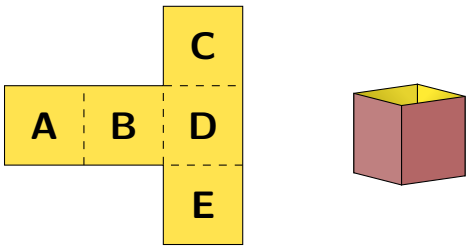
Cut each pizza into thirds, then there are 6 pieces, and each student gets 2 piece. This is two-thirds of a pizza each,

hence (D).

**9.** A piece of card is cut out and labelled as shown in the diagram.

It is folded along the dotted lines to make a box without a top.

Which letter is on the bottom of the box?




(A) A (B) B (C) C (D) D (E) E

Face B is adjacent to A and D in the net, and becomes adjacent to C and E once folded. Since the only face adjacent to all 4 other faces is the bottom of the box, this bottom face must be labelled B,

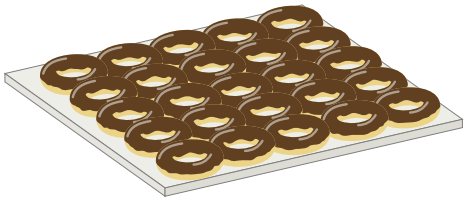
hence (B).

---

**10.** Doughnuts come in bags of 3 and boxes of 8.

I bought exactly 25 doughnuts for my party.

What do I get when I add the number of boxes I bought and the number of bags I bought?



(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

We guess and check the number of boxes of 8 doughnuts.

If there are 0 boxes of 8, all 25 doughnuts are in bags of 3. But 25 is not a multiple of 3 so this is not possible.

If there is 1 box of 8, this leaves $25 - 8 = 17$ doughnuts in bags of 3. Again, this is not possible.

If there are 2 boxes of 8, this leaves $25 - 16 = 9$ doughnuts in bags of 3. So this works if there are 3 bags. Then the total number of bags and boxes is $3 + 2 = 5$.

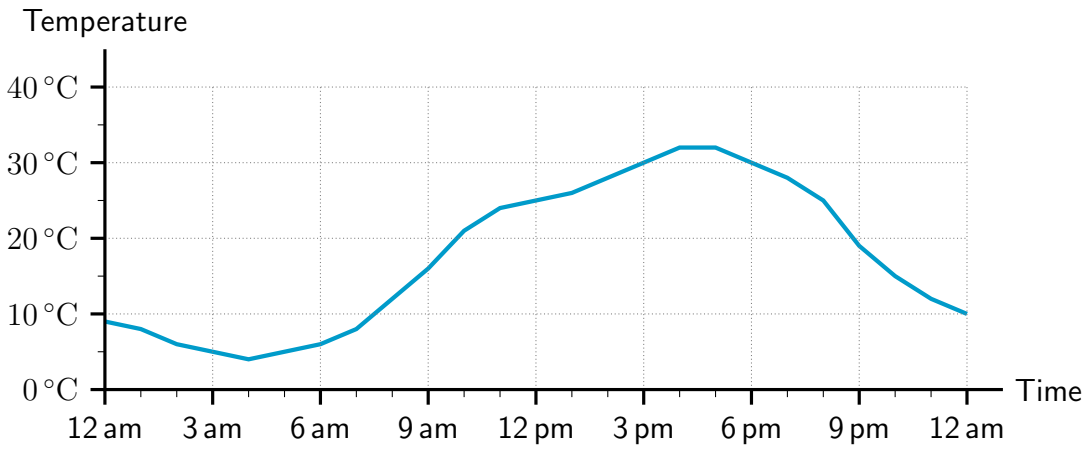
To finish, we check that this is the only possibility: 3 boxes leaves 1 doughnut in a bag, which is impossible, and any more than 3 boxes will contribute more than 25 doughnuts.

So the only possibility is 3 bags and 2 boxes,

hence (B).

## Questions 11 to 20 are worth 4 marks each.

**11.** This line graph shows the temperature each hour during a day.




Roughly for how long was the temperature above 20 °C?

(A) 7 hours (B) 8 hours (C) 9 hours (D) 10 hours (E) 11 hours

The temperature first goes above 20°C at just before 10 am and it next drops below 20°C just before 9 pm. So it is above 20°C for approximately $2 + 9 = 11$ hours,

hence (E).

---

**12.** VIV takes her three children, HANNAH, OTTO and IZZI, out shopping. Each is wearing a t-shirt with their name on the front in capital letters.

When they stand in front of the shop mirror, which names appear the same in the reflection as on the shirts?

(A) VIV and OTTO (B) VIV, OTTO and IZZI

(C) VIV, HANNAH and IZZI (D) HANNAH and OTTO

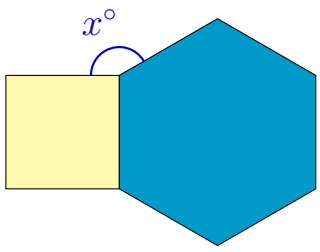
(E) All four of them

All four names are palindromic (reading the same forwards as backwards) but only VIV and OTTO also use symmetric letters. So only VIV and OTTO look the same in the mirror,

hence (A).

---

**13.** This regular hexagon has angles of 120° and the square has angles of 90°.
What is the angle $x°$ in the diagram?




(A) 90° (B) 120° (C) 135° (D) 150° (E) 180°

At the point where the angle $x°$ is marked there are angles of $90° + 120° = 210°$ out of 360°, so $x = 360 - 210 = 150$,

hence (D).

---

**14.** Syed's mother had some money to share with her family.
She gave one-quarter of her money to Syed.
Then she gave one-third of what was left to Ahmed.
Then she gave one-half of what was left to Raiyan.
She was left with \$15, which she kept for herself.
How much money did Syed's mother have to start with?

(A) \$30 (B) \$45 (C) \$60 (D) \$90 (E) \$120

*Alternative 1*

Working backwards, the \$15 she at the end was after she gave half of what she had to Raiyan. So she had \$30 before giving to Raiyan.

This \$30 was after giving one-third to Ahmed, so \$30 is two-thirds of what she had before that, which means she must have given Ahmed \$15 out of the \$45 she had before giving to Ahmed.

This \$45 is after giving one-quarter of her original money to Syed. So \$45 was three-quarters of her original money, meaning that one-quarter was \$15 and her original amount was \$60,

hence (C).

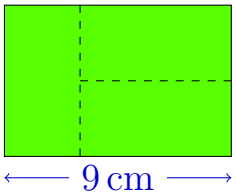
*Alternative 2*

For every \$1 that she started with, she first gives 25 cents to Syed, leaving her with 75 cents. Then 25 cents to Ahmed, leaving her with 50 cents. Then 25 cents to Raiyan, leaving her with 25 cents. But she was left with \$15, with is 60 times as much as this. So she must have started with \$60 and each family member received \$15,

hence (C).

---

**15.** The rectangle shown has a side length of 9 cm. It is divided into 3 identical rectangles as shown. What is the area, in square centimetres, of the original rectangle?




(A) 45 (B) 50 (C) 52 (D) 54 (E) 63

As the rectangle is divided into 3 rectangles the same size, the length of each must be twice the breadth.

Hence the breadth of the original rectangle, must be $\frac{2}{3}\times$ the length which is $\frac{2}{3} \times 9 = 6\,\text{cm}$.

So the width of the original rectangle is 6 cm, and its area is $9 \times 6 = 54$ square centimetres,

hence (D).

**16.** This diagram shows a rectangle with a perimeter of 30 cm. It has been divided by 2 lines into 4 small rectangles. Three of the small rectangles have the perimeters shown. What is the perimeter of fourth small rectangle?

| | |
|---|---|
| Perimeter = 12 cm | Perimeter = 16 cm |
| Perimeter = 14 cm | ? |

(A) 10 cm (B) 12 cm (C) 14 cm (D) 16 cm (E) 18 cm

*Alternative 1*

(AK) Looking at the top two rectangles, the right one has perimeter 4 more than the left one. Since they're the same height, the right one is 2 wider than the left one.

Then in comparison to the bottom-left rectangle the bottom-right rectangle is 2 bigger in width, so 4 bigger in perimeter. So its perimeter is 18,

hence (E).

*Alternative 2*

(AK2) The top-right and the botttom-left rectangles together use the same length line segments as the top-left and the bottom-right rectangles. The total perimeter of the top-left plus the bottom-right rectangles is 30, so the bottom-right rectangle has perimeter $30 - 12 = 18$,

hence (E).

**17.** There are 10 questions in a test. Each correct answer scores 5 points, each wrong answer loses 3 points, and if a question is left blank it scores 0 points.

Tycho did this test and scored 27 points. How many questions did Tycho leave blank?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

---

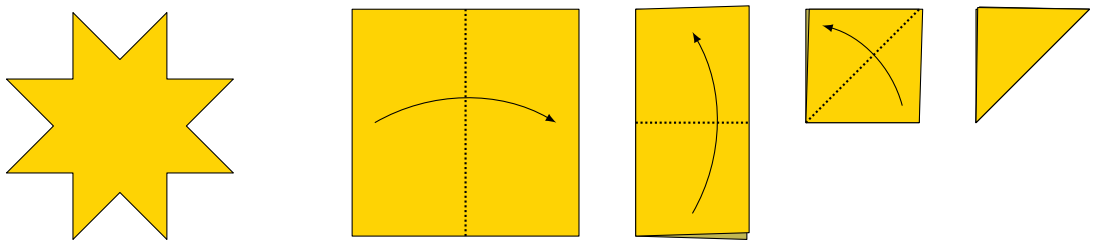
One possibility is that Tycho scores $6 \times 5 = 30$ points from 6 correct questions, then loses 3 points from 1 incorrect question.

For there to be another solution, this solution would have either (i) 3 fewer correct answers and 5 fewer incorrect answers or (ii) 3 more correct answers and 5 more incorrect answers.

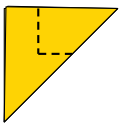
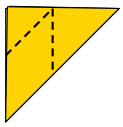
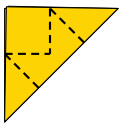
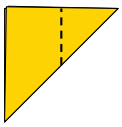
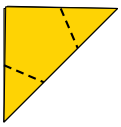
However (i) is not possible, he would have $1 - 5 = -4$ incorrect answers. Also (ii) is not possible, since he would have 9 correct and 6 incorrect answers, out of the 10 questions.

So the only solution has 6 correct, 1 incorrect and 3 blank answers, hence (C).

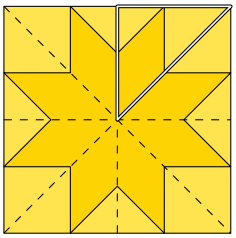
**18.** Estelle is making decorations shaped like the 8-pointed star shown. She folds a square of paper to make a triangle with 8 layers as shown.



How could she cut the triangle so that the unfolded shape is the star?

(A)  (B)  (C)  (D)  (E) 

Draw the page as if it had been unfolded without cutting, with the star on top:



In the outlined triangle, the paper that is part of the star is a rhombus, matching choice (B),

hence (B).

**19.** Earlier this year Ben said, 'Next year I will turn 13, but 2 days ago I was 10.'

Ben's birthday is

(A) 1st January (B) 2nd January (C) 29th December (D) 30th December (E) 31st December

Since Warren's 13th birthday is next year, his 11th was last year. The last possible day he could have been 10 was 30th of December. But the day he made the statement this year was 2 days after a day that he was 10, so the latest possible day he could have said "two days ago I was 10" was the 1st of January this year. So the only possibility is if the day he was 10 was the 30th of December last year, his 11th birthday was the 31st of December, then he made the statement on the 1st of January,

hence (E).

---

**20.** Peyton, Luka and Dan have 180 stickers in total. Peyton has half as many stickers as Luka. Dan has three times as many as Luka.

How many stickers does Peyton have?

(A) 20 (B) 24 (C) 30 (D) 40 (E) 54

For every 1 sticker Peyton has, Luka has 2 and Dan has 6. So there are 9 stickers in total for every 1 sticker Peyton has.

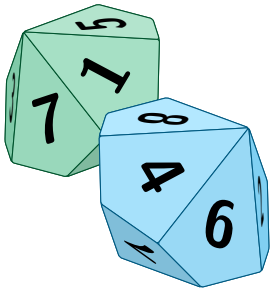
Since there are 180 stickers, Peyton has $180 \div 9 = 20$ stickers,

hence (A).

---

## Questions 21 to 25 are worth 5 marks each.

**21.** Mrs Graaf invents a game for her students to practise arithmetic. They roll two 10-sided dice to pick two random numbers. Starting at one of the numbers, they keep adding the other number until they reach a 3-digit number.




Ian rolls a 5 and an 8. If he chooses to start with 5 and then add 8 again and again, his list is 5, 13, 21, . . . , 93, 101. If he chooses to start with 8 and add 5, his list is 8, 13, 18, . . . , 98, 103.

On Nara's turn, she makes a list that ends with 107.

What pair of numbers could she have rolled?

(A) 4 and 8 (B) 5 and 7 (C) 3 and 4 (D) 6 and 9 (E) 3 and 8

---

*Alternative 1*

Starting with an even number and adding an even number can only produce even numbers. Since 107 is odd, this rules out (A).

Similarly, starting with and adding multiples of 3 can only produce multiples of 3. Since 107 is not a multiple of 3, this rules out (D).

If the adding number is less than 8, then 107 would not be the first 3-digit number in the list. This rules out (B) and (C).

The only remaining possibility is (E), starting with 3 and adding 8. We can check that the list is 3, 11, . . . , 99, 107. Alternatively, we can observe that any number in this list must have a remainder of 3 upon division by 8, and 107 is such a number,

hence (E).

*Alternative 2*

(AK) Since 107 is the first three-digit number in Nara's list, the last addition is either $98 + 9 = 107$ or $99 + 8 = 107$. In each case we backtrack to the starting number.

For the case $98+9 = 107$ the number being added is 9, so we subtract $98 - 10 \times 9 = 8$ to find the starting number. So the numbers rolled could have been 8 and 9.

For $99 + 8 = 107$ we subtract $99 - 12 \times 8 = 3$ so that the starting number is 3, and the two numbers rolled are 3 and 8.

Of the two solutions found, only "3 and 8" is one of the options,

hence (E).

**22.** At a school concert, the tickets cost \$20 per adult and \$2 per child. The total paid by the 100 people who attended was \$920. How many were children?

(A) between 25 and 35 (B) between 35 and 45

(C) between 45 and 55 (D) between 55 and 65

(E) between 65 and 75

*Alternative 1*

(WA) Clearly there must be more children than adults for the total taking to be \$920.

Using a guess and check approach:-

Try 30 adults and 70 children. This gives $30 \times 20 + 70 \times 2 = 740$, not enough.

Try 40 adults.

This gives $40 \times 20 + 60 \times 2 = 920$, so 60 children,

hence (D).

*Alternative 2*

(AK-unitary) If the 100 people were all children, the total paid would have been \$200, which is \$720 less than the actual total.

For every adult, the total raises by \$18. So $720 \div 18 = 40$ adults went, and $100 - 40 = 60$ children,

hence (D).

*Alternative 3*

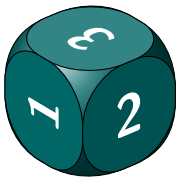
(AK-table) The more adults attend, the higher the total, and there can't be as many as 46 adults. So we test the total amount for different numbers of adults and children, based on the numbers in the question, but only up to 45 adults and 55 children.

| | | | |
|---|---|---|---|
| Adults | 25 | 35 | 45 |
| Price for adults | \$500 | \$700 | \$900 |
| Children | 75 | 65 | 55 |
| Price for children | \$150 | \$130 | \$110 |
| Total price | \$650 | \$830 | \$1010 |

Consequently the total price is \$920 for somewhere between 55 and 65 children,

hence (D).

**23.** Meena has a standard dice, with each pair of opposite faces adding to 7. At first, the three faces she can see add to 6, as shown.



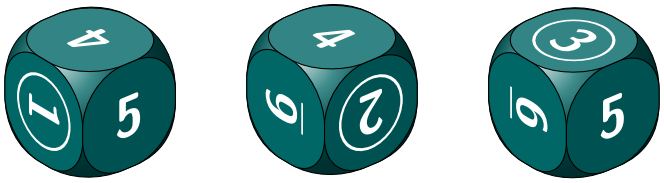

She holds the dice between a pair of opposite faces and rotates it 180°, keeping these opposite faces facing the same direction. She puts the dice back down and adds up the three faces she can now see.

What is the smallest possible total she could get?

(A) 6 (B) 8 (C) 10 (D) 12 (E) 14

---

The pair of faces she holds it by are either $\{1, 6\}$, $\{2, 5\}$ or $\{3, 4\}$. The rotation by 180° keeps this pair in the same position, but changes all other numbers by swapping with the opposite face. So the three possible views of the dice are




Adding visible faces, gives either 10, 12 or 14, and the smallest of these is 10,

hence (C).

**24.** I have 4 whole numbers that add up to 98.

If I were to add 6 to the first number, subtract 6 from the second number, multiply the third number by 6 and divide the fourth number by 6, the four answers would all be the same.

What is the sum of the largest two of my original four numbers?

(A) 72 (B) 86 (C) 88 (D) 90 (E) 94

---

When I divide the largest number by 6, it must still be a multiple of 6 as I multiply the third number by 6, so the largest number must be a multiple of 36.

Try 36,

divide by 36 by 6 and get 6,

This means the others would be 0, 12 and 1, does not work.

Try 72,

divide by 6 and get 12, then the others $2 \times 6 = 12$, $18 - 6 = 12$. and $6 + 6 = 12$, which works, with sum $2 + 6 + 18 + 72 = 98$

So the numbers are 2, 6, 18 and 72.

The sum of the largest two numbers is $18 + 72 = 90$,

hence (D).

---

**25.** When I ride my bike at 20 kilometres per hour, each wheel turns at 2 revolutions per second.

When I ride 1 kilometre, how many revolutions does each wheel make?

(A) 40 (B) 240 (C) 320 (D) 360 (E) 420

---

*Alternative 1*

(WA) In 1 hour, I travel 20 km and the wheels make $2 \times 60 \times 60 = 7200$ revolutions.

So, in 1 kilometre, the wheels revolve $\dfrac{7200}{20} = 360$ times,

hence (D).

*Alternative 2*

(AK) At 20 km per hour, 1 km takes 3 minutes, which is 180 seconds.

At 2 revolutions per second, this is 360 revolutions,

hence (D).

---

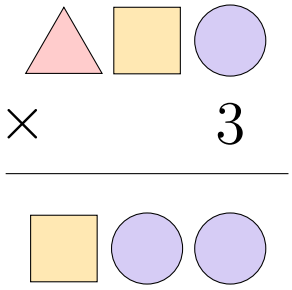
**For questions 26 to 30, colour in the bubbles on the answer sheet to record whole-number answers from 0 to 999.**

**See the back cover for more details.**

**Questions 26 to 30 are worth 6, 7, 8, 9 and 10 marks respectively.**

**26.** In this puzzle, △, □ and ○ represent different nonzero digits.

What is the three-digit number △□○?




There must be ○ = 5 in the units column. Then there is a carry of 1 from the units to the tens column.

In the tens column the last digit of □ × 3 must be 4. That is, □ = 8. Then there is a carry of 2 from the tens to the hundreds column.

In the hundreds column △×3 = 6 so that △ = 2. In all, △□○ = 285, hence (285).

**27.** Li attempted to multiply a single-digit number by 36, but he accidentally multiplied by 63 instead. His answer was 189 larger than the correct answer.

What was the correct answer to the multiplication?

---

*Alternative 1*

The difference between the two answers is the difference between 63 times the digit and 36 times the digit, which is 27 times the digit.

So 27 times the digit is 189, and the digit is $189 \div 27 = 7$. Then the correct multiplication is $7 \times 36 = 252$,

hence (252).

*Alternative 2*

(GV) Let the single-digit number be $x$, and the correct answer be $y$. Then

$$\begin{aligned} 36x &= y \\ 63x &= y + 189 \\ 63x - 36x &= 189 \\ 27x &= 189 \\ 9x &= 189 \div 3 = 63 \\ y = 36x &= 63 \times 4 = 252 \end{aligned}$$

hence (252).

---

**28.** Using 9 out of the 10 possible digits Safia writes 3 numbers, each between 100 and 999.

She adds her 3 numbers together. What is the smallest possible sum?

---

A zero cannot be used in the hundreds place, so to make the sum as small as possible Safia chooses 1, 2 and 3 for the hundreds digits. A zero can now be used in the tens places, so she uses 0, 4 and 5 for the tens digits. For the units digits she uses 6, 7 and 8.

The three numbers can be formed by grouping the chosen hundreds, tens and units digits in various ways, but their sum will always be the same. So, among other possibilities, that sum can be calculated as $106 + 247 + 358 = 711$,

hence (711).

---

**29.** Yifan has a construction set consisting of red, blue and yellow rods. All rods of the same colour are the same length, but differently coloured rods are different lengths. She wants to make quadrilaterals using these rods.

- When she uses two red, one blue and one yellow rod, the perimeter of the quadrilateral is 36 cm.
- When she uses two blue, one red and one yellow rod, the perimeter is 35 cm.
- When she uses two yellow, one blue and one red rod, the perimeter is 33 cm.

What number do you get when you multiply the lengths of one red rod, one blue rod and one yellow rod?

---

(AK) Label the rods R, B and Y.

By comparing the first and second quadrilaterals, R is 1 cm longer than B

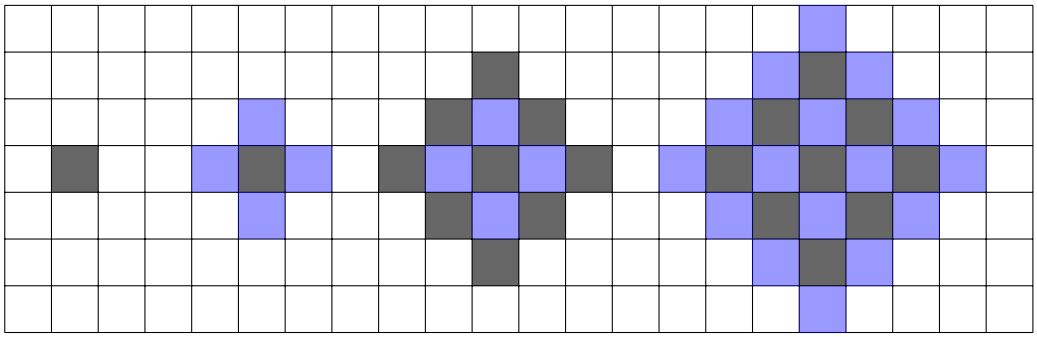
By comparing the first and third quadrilaterals, R is 3cm longer than Y

So a square with 4 R's would have perimeter $36 + 1 + 3 = 40$. Consequently $R = 10$, and then $B = 9$ and $Y = 7$.

Multiplying $R$, $B$ and $Y$ gives $10 \times 9 \times 7 = 630$,

hence (630).

**30.** Janus is making patterns using square tiles. Each pattern is made by copying the previous pattern, then adding a tile to every grid square that shares an edge with the copied pattern.



Pattern 1 Pattern 2 Pattern 3 Pattern 4

His last pattern is the largest one that can be made with fewer than 1000 tiles.

How many tiles are in this last pattern?

---

The shading in the tiles shows that each pattern is found by adding two adjacent square numbers:

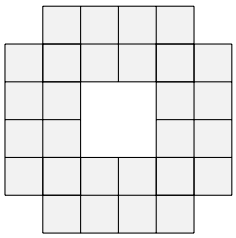
$$0+1=1 \qquad 1+4=5 \qquad 4+9=13 \qquad 9+16=25 \qquad \ldots$$

For this to reach near 1000, the two square numbers will be near 500. Checking, $21^2 = 441$, $22^2 = 484$ and $23^2 = 529$. Then $441+484 = 925$ and $484+529 = 1013$ so that the last pattern that Janus makes has 925 tiles,

hence (925).

# 1 Junior solutions

**1.** Kurt paved his courtyard in the pattern shown. How many $1 \times 1$ pavers are in his courtyard?



(A) 28 (B) 30 (C) 32 (D) 34 (E) 36

Relative to the $6 \times 6$ square, there are $6 \times 6 - 8 = 36 - 8 = 28$ pavers, hence (A).

---

**2.** Which of the following expressions has the smallest value?

(A) $3 + 2$ (B) $3 - 2$ (C) $3 \times 2$ (D) $3 \div 2$ (E) 32

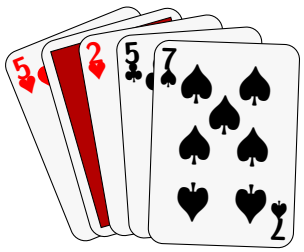
The expression have values:

$$3 + 2 = 5, \quad 3 - 2 = 1, \quad 3 \times 2 = 6, \quad 3 \div 2 = 1\tfrac{1}{2}, \quad 32 = 32.$$

So the expression with the smallest value is $3 - 2 = 1$, hence (B).

---

**3.** The numbers on the top corners of these five playing cards add to 21. What is the number on the hidden card?

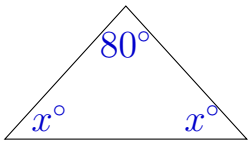


(A) 2 (B) 3 (C) 5 (D) 7 (E) 9

(AK/PC) The cards shown add to $5 + 2 + 5 + 7 = 19$. Then $21 - 19 = 2$ is the number on the hidden card, hence (A).

---

**4.** What is the value of $x$ in this diagram?



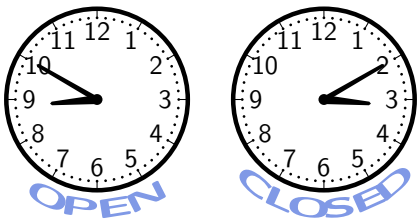

(A) 50 (B) 60 (C) 70 (D) 80 (E) 100

(AK) To make the total of 180° in the triangle, we need $2x = 100$ so that $x = 50$, hence (A).

---

**5.** These two clocks show opening and closing times of my local Marine Discovery Centre.

For how long is it open each day?




(A) 5 hours and 40 minutes

(B) 6 hours and 20 minutes

(C) 6 hours and 40 minutes

(D) 7 hours and 20 minutes

(E) 7 hours and 40 minutes

The time that the centre is open can be counted as 10 minutes, then 6 hours, then 10 minutes, making 6 hours and 20 minutes,

hence (B).

---

**6.** What is the value of $\dfrac{2+3+4}{7+8+9}$?

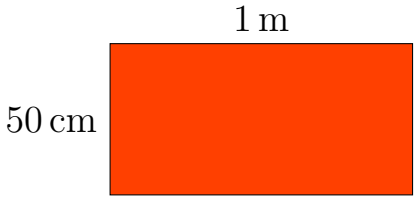
(A) $\dfrac{1}{6}$ (B) $\dfrac{2}{7}$ (C) $\dfrac{3}{8}$ (D) $\dfrac{4}{9}$ (E) $\dfrac{1}{2}$

(PC/GV) $\dfrac{2+3+4}{7+8+9} = \dfrac{9}{24} = \dfrac{3\times 3}{8\times 3} = \dfrac{3}{8}$,

hence (C).

---

**7.** How many $25\,\text{cm} \times 25\,\text{cm}$ squares fit in a $50\,\text{cm} \times 1\,\text{m}$ rectangle?




(A) 1 (B) 2 (C) 4 (D) 6 (E) 8

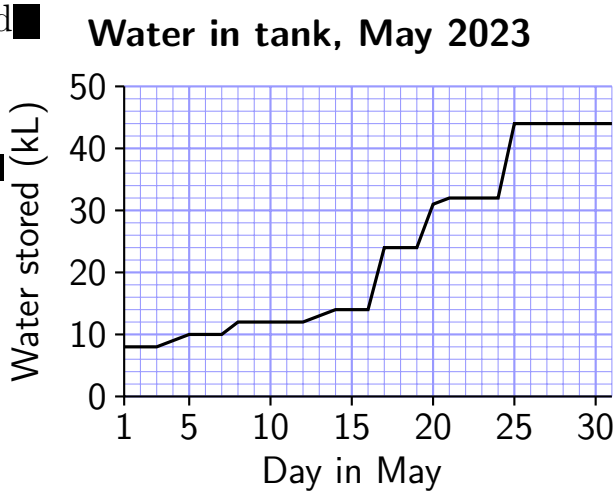
Since $1\,\text{m} = 4 \times 25\,\text{cm}$ and $50\,\text{cm} = 2 \times 25\,\text{cmm}$, the rectangle can be divided into 4 squares horizontally and 2 vertically for a total of $4 \times 2 = 8$ squares,

hence (E).

---

**8.** This graph shows the water stored in my rainwater tank last May. What was the increase in water in the tank between the 10th and the 30th of May?

(A) 32 kL (B) 37 kL (C) 40 kL (D) 42 kL (E) 43 kL




On the 10th of May, the tank had 12 kL. On the 30th of May, it had 44 kL. The increase was $44 - 12 = 32$ kL,

hence (A).

---

**9.** The sum of the ages of Ali, Bey and Cam is 32 years. What will be the sum of their ages in 3 years time?

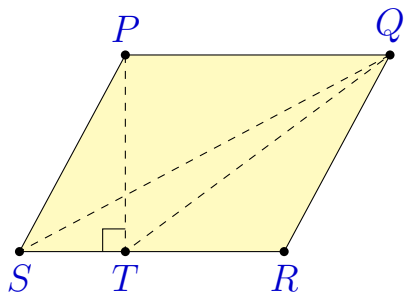
(A) 33 years (B) 35 years (C) 38 years (D) 41 years (E) 96 years

In 3 years, each of the 3 ages will increase by 3 years, so the increased total is $32 + 3 \times 3 = 32 + 9 = 41$ years

---

**10.** A parallelogram $PQRS$ has an area of $60\,\text{cm}^2$ and side $PQ$ of length 10 cm. Which length is 6 cm?

(A) $RQ$ (B) $RS$ (C) $QT$ (D) $PT$ (E) $QS$




(AK) The parallelogram's area is $A = bh$ where $b$ the base and $h$ the height perpendicular to that base. With base $b = PQ = 10$ cm and area $A = 60\,\text{cm}^2$ we must have $h = 6$. Of the given segments, $PT$ is perpendicular $PQ$ so $PT = 6$ cm. All other given segments are longer,

hence (D).

---

**11.** Mei can travel to her grandma's house by a direct route, or by a scenic route that is 5 km longer. When she travels by the scenic route, and comes directly home, the round trip is 35 km. How long is the direct route?

(A) 5 km (B) 12.5 km (C) 15 km (D) 20 km (E) 22.5 km

*Alternative 1*

(JB/GV) Let $s$ be the length of the scenic route and $d$ the direct route. Then $s = d + 5$ and $s + d = 35$.

Then $d + 5 + d = 35$ so that $2d = 30$ and $d = 30 \div 2 = 15$,

hence (C).

*Alternative 2*

(AK/PC) Suppose Mei travels both ways by the direct route. She'll save 5 km, making a round trip of $35 - 5 = 30$ km. So the direct route is $30 \div 2 = 15$ km,

hence (C).

---

**12.** In the game of Nurdle, players try to guess a three-digit number. After each guess, the digits are marked with one of the following three symbols:

| | | |
|---|---|---|
| 5 | 3 | 5 |
| × | ⊙ | × |
| 1 | 2 | 3 |
| × | ✓ | ⊙ |

- ✓ if the digit is correct and in the right place
- ⊙ if the digit is correct, but is not yet in the right place
- × if the digit is not in the three-digit number at all.

Mike's guesses so far are shown.

How many different three-digit numbers fit the clues he has so far?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

/AK/GV/PC) Digits 1 and 5 are not in the hidden number, and the second digit is 2. There is a 3, but neither the 2nd nor 3rd digits are 3, so the first digit is 3. So far, the number is $\boxed{3}\boxed{2}\boxed{?}$, where the last digit can't be 1, 3 or 5. The other 7 choices for the last digit are 0, 2, 4, 6, 7, 8, 9, which are all possible,

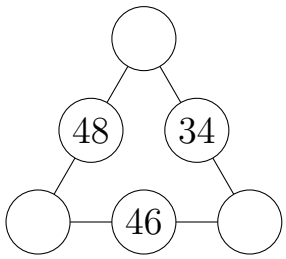
hence (C).

---

**13.** In this triangle, the number in the centre of each side is the sum of the two numbers at the ends of the sides.

What is the sum of the three numbers at the ends of the sides?

(A) 46 (B) 52 (C) 59 (D) 64 (E) 68




*Alternative 1*

(WA/PC/GV) The three numbers from the three sides add to $48+46+34=128$. In this total, each of the three corner numbers is counted twice, so the sum of the end numbers is $128 \div 2 = 64$,

hence (D).

*Alternative 2*

Due to the 48 and the 46, the top number is 2 more than the bottom right number. So instead of $17+17=34$ we use $18+16$, so that the three missing numbers are 18, 16 and 20, which add to 64,

hence (D).

---

**14.** A fuel tank is 40% empty. Then 40 litres of fuel is removed. The tank is now 40% full. How many litres are in a full tank?

(A) 40 (B) 100 (C) 160 (D) 200 (E) 400

*Alternative 1*

(WA2/GV) Let the full volume be $v$. Then

$$\begin{aligned} \frac{60}{100}v &= \frac{40}{100}v + 40 \\ 60v &= 40v + 4000 \\ 20v &= 4000 \\ v &= 200, \end{aligned}$$

hence (D).

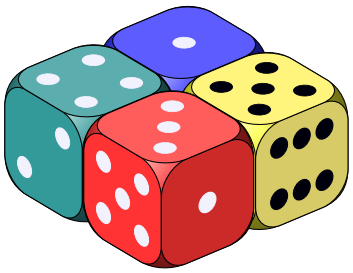
*Alternative 2*

(AK/PC) The tank is initially 40% empty, which is 60% full, and then it becomes 40% full. So the 40 L removed is 20% of the tank. Then $5 \times 40 = 200$ L is 100% of the tank,

hence (D).

---

**15.** The game *Tutu* uses 4 standard dice with opposite faces adding to 7. The dice are rolled and randomly arranged into a $2 \times 2$ block.



A block's score is the sum of the 4 numbers showing on top plus the sum of the 8 numbers showing around the outside.

What is the largest possible score for the block shown?

(A) 44 (B) 46 (C) 48 (D) 50 (E) 52

---

*Alternative 1*

The total of the eight visible sides is $4+1+3+5+2+5+1+6 = 27$.

The green dice on the left has one more face that contributes to the sum. Since 2 and 4 are showing, the third face could be a 6.

The yellow dice on the right also has one more face that contributes. Since 5 and 6 are already showing, the third face is at most 4.

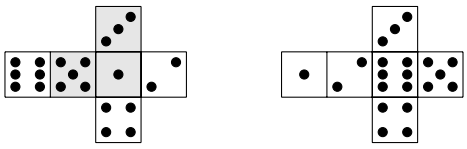
The blue dice at the back is showing a 1. This means that 6 is on the bottom and doesn't contribute to the sum. But 4 and 5 are available.

So the hidden faces have sum at most $6+4+4+5 = 19$ and the total is at most $19 + 27 = 46$,

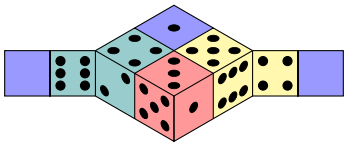
hence (B).

*Alternative 2*

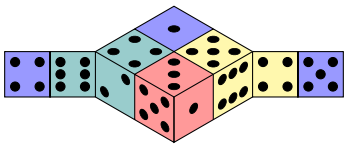
(AK) Here is a net for the red dice, based on the three faces visible, and a second net with cuts chosen to help with the green and yellow dice:



Comparing this net to the green and yellow dice, we can carefully identify two of the hidden faces in the block, shown here partially opened out:



This leaves the two back faces of the blue dice, both adjacent to the top face. From the first net, the largest sum possible is $5 + 4 = 9$. Here is the complete solution:



The visible faces then add to 46,

hence (B).

**16.** The five clocks below are all showing the wrong time. One clock is 3 minutes away from the correct time and another is 7 minutes away. Which clock is closest to the correct time?

(A)

(B)

(C)

(D)

(E)


*Alternative 1*

Consider the clocks that are out by 3 and 7 minutes.

If one is fast and the other is slow, then the times they show must differ by 10 minutes. But the last digit on each clock is different, so there is no such pair of times among the five shown. So they are either both fast or both slow, and either way their times differ by 4 minutes. The only possibilities are clocks (C) and (D), showing 10:00 and 10:04, respectively.

If clocks (C) and (D) are both slow, then the real time is 10:07. But this is impossible since we know that clock (E) 10:07, is also wrong. So clocks (C) and (D) are both fast and the real time is 9:57. Of the five times shown, 9:58 on clock (B) is closest to being correct,

hence (B).

*Alternative 2*

(AK/GV) For one clock to be out by 3 minutes, possible times are found by adding or subtracting 3 minutes from the given times:

| | | | | |
|---|---|---|---|---|
| 9:52 | 9:55 | 9:57 | 10:01 | 10:04 |
| 9:58 | 10:01 | 10:03 | 10:07 | 10:10 |

Similarly, for another clock to be out by 7 minutes, possible times are

| | | | | |
|---|---|---|---|---|
| 9:48 | 9:51 | 9:53 | 9:57 | 10:00 |
| 10:02 | 10:05 | 10:07 | 10:11 | 10:14 |

Only 9:57 and 10:07 are common to these lists. However, clock (E) shows 10:07 is wrong, which rules out that time. So the time is 9:57, and the closest clock to this is 9:58,

hence (B).

**17.** I wrote a list of all multiples of 7 between 1 and 500. Then I looked at the last digit of each number in my list.

Which digit appeared most often as the last digit?

(A) 1 (B) 3 (C) 7 (D) 9

(E) None of (A)–(D), because there is no single last digit that appears most often

---

*Alternative 1*

ND1/GV/PC) When the multiples of 7 are written out, the units digit repeats every ten numbers.

$$7, 4, 1, 8, 5, 2, 9, 6, 3, 0, 7, 4, 1, 8, 5, 2, 9, 6, 3, 0, \ldots$$

So when the multiples of 7 up to 490 are written out, each digit appears the same number of times as the units digit. There is only one more multiple of 7 that is less than 500 — namely, 497. Therefore, 7 appears the largest number of times as the units digit.

*Alternative 2*

(ND2) If two multiples of 7 have the same units digit, then the difference between them is a multiple of 7 and also a multiple of 10, so this difference is a multiple of 70. Conversely, adding 70 to a multiple of 7 creates another multiple of 7 with the same units digit. So for any units digit, the multiples of 7 (if any) are in a sequence increasing by 70.

For example, 7, 77, 147, ..., 497 are the multiples of 7 ending in 7, and there are 8 of these less than 500. For any other last digit, there will be only 7, since the first multiple of 7 with this digit will be larger than 10, and so the 8th one will be larger than $10 + 490 = 500$. Thus the most frequent last digit is 7,

hence (C).

*Alternative 3*

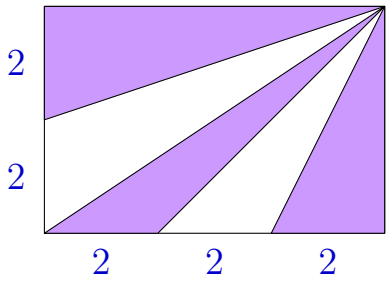
(AK) Since $500 \div 7 = 71\frac{3}{7}$ the list has $7n$ for $n = 1, \ldots, 71$. There is a one-to-one correspondence between last digit of $n$ and last digit of $7n$, as in this table:

| last digit of $n$: | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| last digit of $7n$: | 0 | 7 | 4 | 1 | 8 | 5 | 2 | 9 | 6 | 3 |

But for $n = 1, \ldots, 71$, the most common last digit is 1. So the most common last digit of $7n$ is 7,

hence (C).

**18.** Amy designed this rectangular flag for her fleet of yachts. What fraction of the flag is shaded?




(A) $\frac{2}{3}$ (B) $\frac{3}{5}$ (C) $\frac{5}{8}$ (D) $\frac{1}{2}$ (E) $\frac{7}{12}$

*Alternative 1*

(PC/AK) The flag has area $4 \times 6 = 24$ square units.

The areas, in square units, of three shaded triangles are $\frac{1}{2} \times 2 \times 6 = 6$, $\frac{1}{2} \times 2 \times 4 = 4$ and $\frac{1}{2} \times 2 \times 4 = 4$, from left to right.

So $6+4+4 = 14$ square units are shaded, which is $\frac{14}{24} = \frac{7}{12}$ of the flag, hence (E).

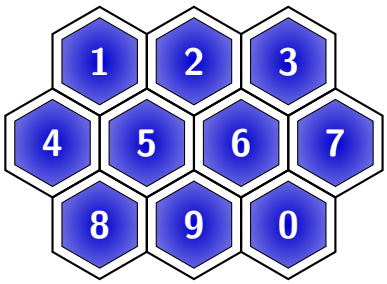
*Alternative 2*

(GV/AK) Consider the two white triangles. The top one can be formed by halving the area of the rectangle (along the diagonal) and then halving the area of the resulting triangle (along a median). So it is $\frac{1}{4}$ of the rectangle.

Similarly, the bottom white triangle is one-third of one half of the rectangle, which is $\frac{1}{6}$. So the total white area is $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$ of the rectangle and the shaded area is $\frac{7}{12}$ of the rectangle,

hence (E).

**19.** To unlock a security door, Daniyal enters a code on the keypad shown. To make it easy to remember:




- the code has 10 digits
- the code uses every digit exactly once
- the code steps from button to button only if the two buttons touch
- the code is as large as possible.

What is the 5th digit of Daniyal's code?

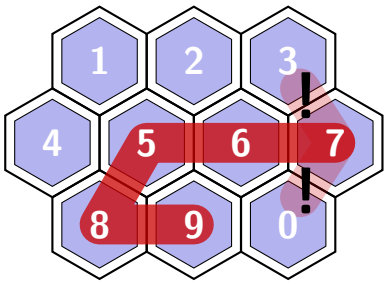
(A) 0 (B) 1 (C) 2 (D) 6 (E) 7

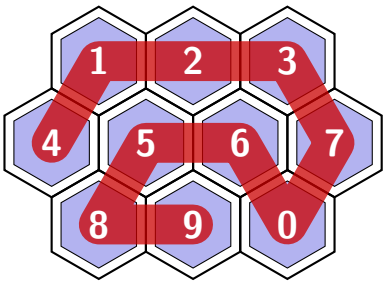
---

*Alternative 1*

To make the largest number we start with digit 9. We then continue to the next highest adjacent button for as long as this doesn't cut off any other buttons on the keypad. So the first four digits of the code are 9856.

If we try to go to 7 next, this will cut off the 0 or the other four buttons, as shown on the left, and the code cannot be completed. Similar issues arise if we try to go to 3 or 2 next.




So the next button must be 0 and the rest of the code can be completed as shown on the right,

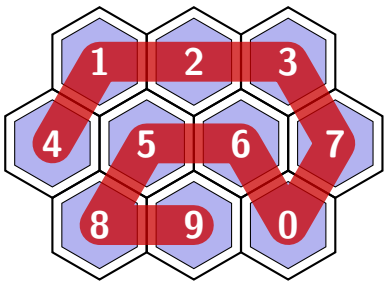
hence (A).

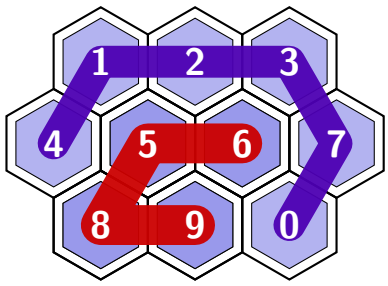
*Alternative 2*

(CW/AK) By starting at 9 and always choosing the largest neighbouring button, the path 985673214 can be found. However this path misses button 0, so the modified path 9856073214, shown below left, seems likely to be the best (largest) solution.

To confirm that 9856073214 is best, notice that no solution can have a better first 4 digits than 9856. Once we have used 9856, the rest

of the path must string together the remaining six buttons, either as 073214 (possible) or 412370 (impossible).




The only complete path is then 9856073214, which must be the best, hence (A).

---

**20.** Jake had 3 times as many gold balloons as blue balloons, and no other colours. Then some blew away, leaving him with 5 times as many blue balloons as gold balloons.

What is the smallest number of balloons that Jake could have started with?

(A) 15 (B) 20 (C) 24 (D) 30 (E) 100

*Alternative 1*

Let $x$ be a number of gold balloons at the end. Then the number of blue balloons will be $5x$ at the end. Hence, there were at least $3 \times 5x = 15x$ gold balloons at the beginning and at least $20x$ all balloons in total. Then, the least total number of balloons can be achieved for $x = 1$ which is 20 balloons.,

hence (B).

*Alternative 2*

(AK) Jake ended with at least 5 blue balloons, so he must have started with at least 5 blue balloons and at least 15 gold balloons. So he started with at least 20 balloons.

Also, there is a solution with exactly 20 balloons, starting with 5 blue and 15 gold and ending with 5 blue and 1 gold,

hence (B).

---

**21.** Li attempted to multiply a single-digit number by 36, but he accidentally multiplied by 63 instead. His answer was 189 larger than the correct answer.

What was the correct answer to the multiplication?

(A) 180 (B) 189 (C) 216 (D) 252 (E) 288

---

*Alternative 1*

The difference between the two answers is the difference between 63 times the digit and 36 times the digit, which is 27 times the digit.

So 27 times the digit is 189, and the digit is $189 \div 27 = 7$. Then the correct multiplication is $7 \times 36 = 252$,

hence (D).

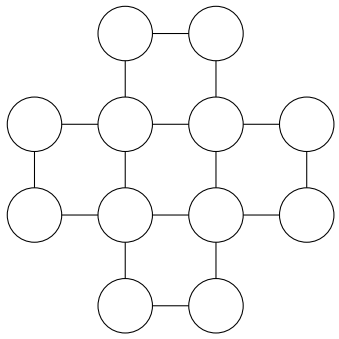
*Alternative 2*

(GV) Let the single-digit number be $x$, and the correct answer be $y$. Then

$$\begin{aligned} 36x &= y \\ 63x &= y + 189 \\ 63x - 36x &= 189 \\ 27x &= 189 \\ 9x &= 189 \div 3 = 63 \\ y = 36x &= 63 \times 4 = 252 \end{aligned}$$

hence (D).

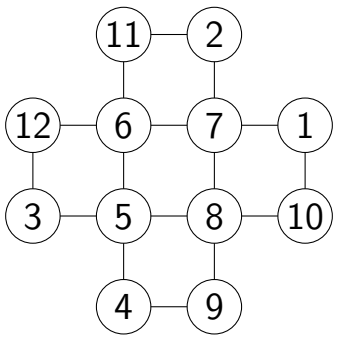
**22.** The numbers from 1 to 12 are to be placed on the vertices of the 5 interconnected squares in this diagram. The sum of the 4 vertices of any of the 5 squares is a constant. What is this constant?



(A) 26 (B) 28 (C) 30 (D) 32 (E) 34

(MC) The numbers from 1 to 12 sum to 78. Let the constant be $x$. If we combine all 5 squares into one sum, the numbers around the middle square will be counted 3 times, as each vertex is on three different squares. Hence $78 + 2x = 5x$ giving $x = 26$.

With this constant, the numbers can be placed in many ways, for instance:




hence (A).

**23.** Ann is running round her favourite running course at a steady pace of 5 minutes per kilometre. She passes a marker which she knows is one-quarter of the way round. One minute and 40 seconds later, she passes another marker which is one-third of the way round. How long will it take her to run the whole course?

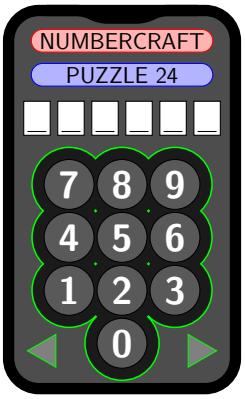
(A) 18 minutes (B) 20 minutes (C) 20 minutes and 40 seconds (D) 22 minutes and 30 seconds (E) 24 minutes

(MC/GV) In the 100 seconds between passing the two markers, Ann completes $\frac{1}{3} - \frac{1}{4} = \frac{1}{12}$ of the course. So she will take $12 \times 100 = 1200$ seconds to complete her run, which is 20 minutes,

hence (B).

---

**24.** Jaz unlocked a secret level in the game Numbercraft. She entered a 6-digit code where the sum of the first three digits equals the sum of the last three digits. Further, she gained a bonus, since her code had four 7s in a row.




For instance, she could have entered 077770.

How many different codes could she have entered to get the secret level and the bonus?

(A) 10 (B) 18 (C) 20 (D) 27 (E) 64

(OY/GV) If the code has 7777 in the middle, the first and last digits are the same, so there are 10 such codes: 077770, 177771, . . . , 977779.

If the codes starts 7777, the last two digits must add to 14, so they are 59, 68, 77, 86 or 95. Since 777777 was found in the first case, there are 4 additional codes.

Similarly there are 4 additional codes that end 7777. Consequently there are $10 + 4 + 4 = 18$ codes that give Jaz both the secret level and the bonus,

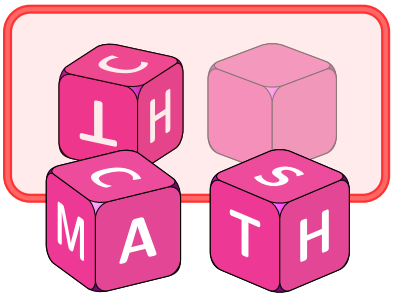
hence (B).

---

**25.** Two identical dice have faces labelled A, C, H, M, S and T.

The diagram shows what they look like when placed in front of a mirror, but the reflection of the second dice has been left blank.

What does the reflection of the second dice look like?



(A) (B) (C) (D) (E)

(AK/GV) The top face is correct in all five answers, so we don't need to consider it further. Using the left dice and its reflection, we can identify some opposite faces: A and T are opposite, and M and H are opposite. So on the right reflection, the vertical faces are either MA or MA. However, A and M are up the same way on the left dice, and only MA has them up the same way on the right reflection, which is option (E),

hence (E).

---

**26.** I start with a number, multiply it by 10, and then subtract a multiple of 9 that is less than 100. My answer is 5347. What number did I start with?

(WA/AK) Reversing the process, when we add 5347 and the multiple of 9 we get 10 times the original number, with units digit 0. So the multiple of 9 must have units digit 3. The only multiple of 9 less than 100 with units digit 3 is 63. Since $5347 + 63 = 5410$ the original number must have been 541,

hence (541).

---

**27.** Amelia noticed that the names of three friends, Mei, Emma and Liam, were all made from letters of 'Amelia'. She chose different values from 0 to 9 for each of $A$, $E$, $I$, $L$ and $M$ so that

$$M + E + I \;=\; E + M + M + A \;=\; L + I + A + M$$

What is the largest possible value of $A + M + E + L + I + A$?

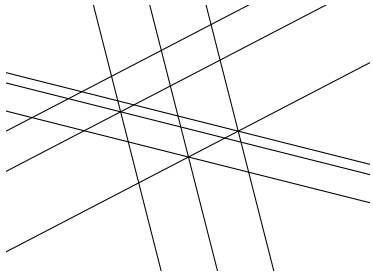
(AK/GV)

$$\begin{aligned} M + E + I &= E + M + M + A \\ \implies I &= A + M \\ E + M + M + A &= L + I + A + M \\ \implies E &= L + I - M = L + A \\ A + M + E + L + I + A &= (A + M) + (L + A) + E + I \\ &= 2E + 2I \end{aligned}$$

Since $E + I$ is at most $8 + 9 = 17$, $A + M + E + L + I + A$ is at most 34. Setting $E = 9$, $I = 8$, $A = 2$, $L = 7$ and $M = 6$ gives a solution with $A + M + E + L + I + A = 34$,

hence (34).

**28.** There are three sets of three parallel lines in a plane. Lines in different sets are not parallel, and every pair of non-parallel lines intersect. The diagram shows one way to do this, but for other arrangements the number of points of intersection varies.



Find the largest possible number of intersection points and the smallest possible number of intersection points, then multiply these two numbers.

---

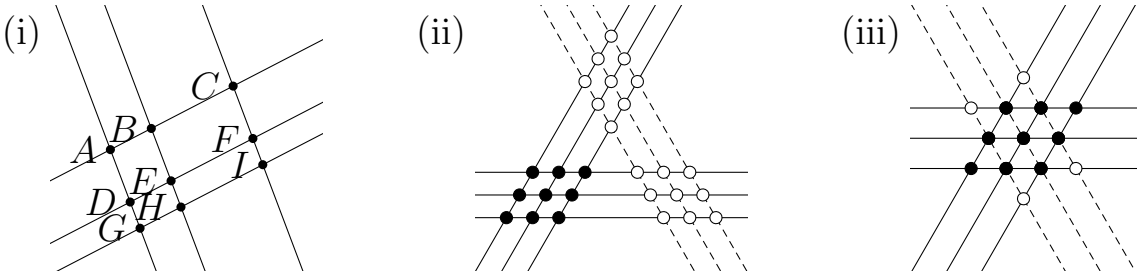
*Alternative 1*

(MC) If there are no common points of intersection between any three lines, then there will be $3 \times 3 \times 3 = 27$ points of intersection. The largest number of common points of intersection of all three lines is 7, which will give $27 - 2 \times 7 = 13$ as the smallest number of intersections. The product is $27 \times 13 = 351$,

hence (351).

*Alternative 2*

(AK2) Draw the first two triples of lines, creating 9 points of intersection as shown in diagram (i). Call these *old* lines and intersections.




The remaining *new* lines create *new* intersection points where a new line crosses exactly one old line. By trial-and-error we can find arrangements with 18 and 4 new intersections, such as in diagrams (ii) and (iii). We claim that these are the maximum and minimum, giving 27 and 13 intersections respectively.

To confirm these claims, consider a single new line. It can include 0, 1, 2 or 3 old intersections, accounting for crossing 0, 2, 4 or 6 old lines respectively. The remaining 6, 4, 2 or 0 crossings (respectively) give new intersections.

So with 3 new lines, there are at most $3 \times 6 = 18$ new intersections, confirming the claim that $9 + 18 = 27$ is the maximum number of intersections.

For the least number of new intersections, there are only two ways that a single new line can include 3 old intersections: $AEI$ and $CEG$. Since these are not parallel, our three new lines can only include one of them. That is, at most one new line can have 3 old intersections. Thus at most $3 + 2 + 2$ old intersections can be included on the new lines. As diagram (iii) shows, this corresponds to $0 + 2 + 2 = 4$ new intersections, and this is the fewest possible. This confirms the claim that $9 + 4 = 13$ is the minimum number of intersections.

Then the product of largest and smallest possibilities is $27 \times 13 = 351$, hence (351).

---

**29.** A two-digit number has the property that when it is divided by the sum of its digits the result is 4 with remainder 3. What is the sum of all two-digit numbers with this property?

(WA/PC) Let the tens digit of the number be $a$ and the units digit be $b$.

Then $10a + b = 4(a + b) + 3$ which simplifies to $b = 2a - 1$.

Checking possibilities

| $a$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| $b$ | 1 | 3 | 5 | 7 | 9 | - | - | - | - |

This gives 11, 23, 35, 47 and 59. However, 11 is not one of the numbers, since $11 \div 2$ doesn't have remainder 3. Each of the other numbers do work.

So, the sum of the numbers with the property is $23+35+47+59 = 164$, hence (164).

---

**30.** Donna likes all numbers that are not divisible by 3 and Sandra likes all numbers that have no digits divisible by 3. How many four-digit numbers are there that both Donna and Sandra like?

(OY/AK) The digits in numbers that Sandra likes are 1, 2, 4, 5, 7, 8, so that she likes $6^4 = 1296$ numbers. Call these *Sandra numbers*. For Donna to like a number, the sum of digits cannot be divisible by 3, due to the test for divisibility by 3. That is, when divided by 3 the remainder is 1 or 2.

Since all the digits have remainder 1 or 2 when divided by 3, we can *reduce* each Sandra number by replacing each digit with the remainder on dividing by 3:

$$1, 4, 7 \longrightarrow 1 \qquad 2, 5, 8 \longrightarrow 2$$
$$\text{e.g. } 1874 \longrightarrow 1211$$

This reduction does not change the remainder on division by 3. For instance both 1874 and 1211 have remainder 2 when divided by 3.

There are $2^4 = 16$ different reduced numbers (listed below) and each reduced number has $3^4 = 81$ Sandra numbers that reduce to it. Further, the remainder on dividing a reduced number by 3 depends only on how many 1s it has. So we classify by how many 1s are in the 16 reduced numbers:

| Reduced numbers | 1111 | 1112 1121 1211 2111 | 1122 1212 1221 2112 2121 2211 | 1222 2122 2212 2221 | 2222 |
|---|---|---|---|---|---|
| # reduced nums | 1 | 4 | 6 | 4 | 1 |
| # Sandra nums | 81 | $4 \times 81 = 324$ | $6 \times 81 = 486$ | $4 \times 81 = 324$ | 81 |
| Rem. when $\div 3$ | 1 | 2 | 0 | 1 | 2 |
| Donna likes? | yes | yes | no | yes | yes |

From this table, the number of Sandra numbers that Donna likes can either be calculated as $(1+4+4+1) \times 81 = 810$ or $1296 - 486 = 810$, hence (810).

## 2 Intermediate solutions

**1.** A 40-minute lesson started at 10:50 am. Exactly half-way through the lesson the fire alarm went off. At what time did the fire alarm go off?

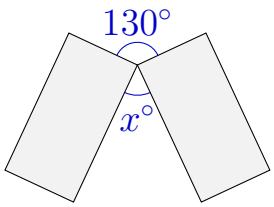
(A) 10:30 am (B) 11:00 am (C) 11:10 am (D) 11:20 am (E) 11:30 am

(AH) Half of the 40-minute lesson is 20 minutes.
Then 20 minutes after 10:50 am is 11:10 am,

hence (C).

---

**2.** Two rectangles have a vertex in common, as shown. What is the size of the angle marked $x°$ between them?




(A) 10° (B) 20° (C) 30° (D) 40° (E) 50°

(AK/GV) The angles around the point where the two rectangles meet are 90°, 90°, 130° and $x°$, so that $x = 360 - (90 + 90 + 130) = 360 - 310 = 50$,

hence (E).

---

**3.** What is the value of $\dfrac{2+3+4}{7+8+9}$?

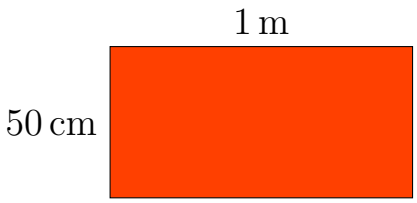
(A) $\dfrac{1}{6}$ (B) $\dfrac{2}{7}$ (C) $\dfrac{3}{8}$ (D) $\dfrac{4}{9}$ (E) $\dfrac{1}{2}$

(PC/GV) $\dfrac{2+3+4}{7+8+9} = \dfrac{9}{24} = \dfrac{3\times 3}{8\times 3} = \dfrac{3}{8}$,

hence (C).

---

**4.** How many 25 cm × 25 cm squares fit in a 50 cm × 1 m rectangle?




(A) 1 (B) 2 (C) 4 (D) 6 (E) 8

Since 1 m = 4 × 25 cm and 50 cm = 2 × 25 cmm, the rectangle can be divided into 4 squares horizontally and 2 vertically for a total of $4 \times 2 = 8$ squares,

hence (E).

---

**5.** Which one of these is equal to $57 \times 953$?

(A) 321 (B) 4321 (C) 54 321 (D) 654 321 (E) 7 654 321

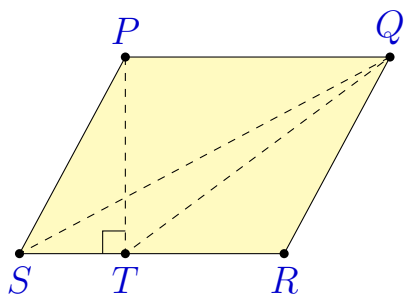
Rounding up, $57 \times 953 < 60 \times 1000 = 60\,000$, which suggests (C). Rounding down, $57 \times 953 > 50 \times 900 = 45\,000$, so that only (C) is possible. Evaluating precisely confirms that $57 \times 953 = 54321$,

hence (C).

---

**6.** A parallelogram $PQRS$ has an area of $60\,\text{cm}^2$ and side $PQ$ of length 10 cm. Which length is 6 cm?

(A) $RQ$ (B) $RS$ (C) $QT$ (D) $PT$ (E) $QS$




(AK) The parallelogram's area is $A = bh$ where $b$ the base and $h$ the height perpendicular to that base. With base $b = PQ = 10\,\text{cm}$ and area $A = 60\,\text{cm}^2$ we must have $h = 6$. Of the given segments, $PT$ is perpendicular $PQ$ so $PT = 6\,\text{cm}$. All other given segments are longer,

hence (D).

---

**7.** Mei can travel to her grandma's house by a direct route, or by a scenic route that is 5 km longer. When she travels by the scenic route, and comes directly home, the round trip is 35 km. How long is the direct route?

(A) 5 km (B) 12.5 km (C) 15 km (D) 20 km (E) 22.5 km

*Alternative 1*

(JB/GV) Let $s$ be the length of the scenic route and $d$ the direct route. Then $s = d + 5$ and $s + d = 35$.

Then $d + 5 + d = 35$ so that $2d = 30$ and $d = 30 \div 2 = 15$,

hence (C).

*Alternative 2*

(AK/PC) Suppose Mei travels both ways by the direct route. She'll save 5 km, making a round trip of $35 - 5 = 30$ km. So the direct route is $30 \div 2 = 15$ km,

hence (C).

---

**8.** What is the value of $((2^0)^2)^3$ ?

(A) 1 (B) 12 (C) 32 (D) 64 (E) 256

*Alternative 1*

(KH) Since $2^0 = 1$ we write $((2^0)^2)^3 = ((1)^2)^3 = (1)^3 = 1$,

hence (A).

*Alternative 2*

(AK) Using the rule $(a^n)^m = a^{mn}$ twice,

$$((2^0)^2)^3 = ((2^{0\times 2})^3 = 2^{0\times 2\times 3} = 2^0 = 1$$

hence (A).

---

**9.** What must 0.05 be divided by to get 50?

(A) 1000 (B) 100 (C) 0.1 (D) 0.001 (E) 0.0001

*Alternative 1*

(AH/PC) Let $x$ be the unknown divisor. Then $\frac{0.05}{x} = 50 \implies 50x = 0.05 \implies x = \frac{0.05}{50} = \frac{0.01}{10} = 0.001$,

hence (D).

*Alternative 2*

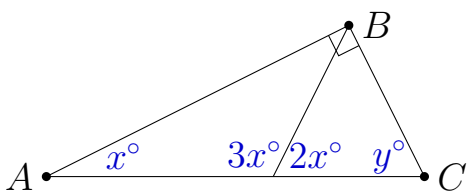
Invert the division question to an equivalent multiplication problem: what must 50 be multiplied by to give 0.05? The answer to this is $0.05 \div 50 = 0.001$,

hence (D).

---

**10.** In the right-angled triangle $ABC$ shown, what is the value of $y$?

(A) 45 (B) 48 (C) 54 (D) 60 (E) 72




GV/PC/DDC) The angles marked $3x^\circ$ and $2x^\circ$ are supplementary, so that $180 = 3x + 2x = 5x$, giving $x = 36$. Then in $\triangle ABC$, $36 + 90 + y = 180$ giving $y = 180 - 126 = 54$,

hence (C).

---

**11.** The number 11 can be written as the sum of three positive whole numbers in many ways. In how many ways can this be done where the numbers are different and in increasing order?

(A) 3 (B) 4 (C) 5 (D) 6 (E) 7

DC/GV/PC) There are 3 possibilities that start with 1: $11 = 1+2+8 = 1+3+7 = 1+4+6$.

There are 2 possibilities that start with 2: $11 = 2+3+6 = 2+4+5$.

Beyond that, the total is at least $3+4+5 = 12$, which is too large.

So there are $3+2 = 5$ possibilities,

hence (C).

---

**12.** A two-digit number is reversed then added to itself. The answer **cannot** be

(A) 55 (B) 110 (C) 132 (D) 154 (E) 186

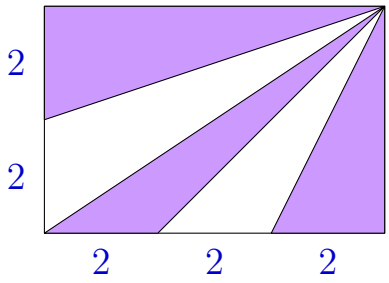
(JB) In the two numbers being added, exchange the units digits. For instance $14+41 = 11+44$. This keeps the sum the same, but now it is clear that the number must be a multiple of 11. Of the given numbers, 186 is not a multiple of 11, so it can't be the sum.

The others can also be checked $55 = 14+41$, $110 = 19+91$, $132 = 66+66$, and $154 = 68+86$ which confirms that only 186 can't be the total,

hence (E).

---

**13.** Amy designed this rectangular flag for her fleet of yachts. What fraction of the flag is shaded?




(A) $\frac{2}{3}$ (B) $\frac{3}{5}$ (C) $\frac{5}{8}$ (D) $\frac{1}{2}$ (E) $\frac{7}{12}$

*Alternative 1*

(PC/AK) The flag has area $4 \times 6 = 24$ square units.

The areas, in square units, of three shaded triangles are $\frac{1}{2} \times 2 \times 6 = 6$, $\frac{1}{2} \times 2 \times 4 = 4$ and $\frac{1}{2} \times 2 \times 4 = 4$, from left to right.

So $6+4+4 = 14$ square units are shaded, which is $\frac{14}{24} = \frac{7}{12}$ of the flag, hence (E).

*Alternative 2*

(GV/AK) Consider the two white triangles. The top one can be formed by halving the area of the rectangle (along the diagonal) and then halving the area of the resulting triangle (along a median). So it is $\frac{1}{4}$ of the rectangle.

Similarly, the bottom white triangle is one-third of one half of the rectangle, which is $\frac{1}{6}$. So the total white area is $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$ of the rectangle and the shaded area is $\frac{7}{12}$ of the rectangle,

hence (E).

**14.** What is the largest possible whole-number value of the expression $a \times b + \frac{c}{d} - \frac{e}{f}$ where $a, b, c, d, e, f$ are the numbers 1, 2, 3, 4, 5 and 6 in some order?

(A) 30 (B) 31 (C) 32 (D) 33 (E) 34

---

*Alternative 1*

(SB/GV) Since multiplication produces the largest possible result, $ab$ should be $5 \times 6 = 30$. Furthermore, we want to maximize $\frac{c}{d}$, so $d = 1$. Unfortunately, if we let $c = 4$, then $\frac{e}{f}$ will not be a whole number. Thus $c = 3$, $e = 4$, $f = 2$, and $ab + \frac{c}{d} - \frac{e}{f} = 30 + \frac{3}{1} - \frac{4}{2} = 31$,

hence (B).

*Alternative 2*

(ineq) Let $N = ab + \frac{c}{d} - \frac{e}{f}$. We first try $ab = 6 \times 5 = 30$. With this, $c, d, e, f$ are 1,2,3,4 (in some order) such that $\frac{c}{d} - \frac{e}{f}$ is an integer. Neither $d$ nor $f$ can be 3 or 4, and we can't use $\frac{3}{2}$, so the two fractions must be $\frac{3}{1} = 3$ and $\frac{4}{2} = 2$. Consequently when $ab = 30$, the largest possible value is $N = 30 + \frac{3}{1} - \frac{4}{2} = 31$.

Next consider $ab < 30$. The next-largest possibility for $ab$ is $ab = 6 \times 4 = 24$, so $ab \leq 24$. However, $\frac{c}{d} - \frac{e}{f} < \frac{c}{d} \leq \frac{6}{1}$, so that $N = ab + \frac{c}{d} - \frac{e}{f} < 24 + 6 = 30$. So there are no solutions larger than the solution $N = 31$ above,

hence (B).

**15.** Four children named, from youngest to oldest, Abdul, Bipin, Cai and Denise have ages which are equally spaced apart. Abdul and Bipin's ages add to 18, whilst Cai and Denise's ages add to 34. How old is Denise?

(A) 14 (B) 16 (C) 18 (D) 19 (E) 20

*Alternative 1*

IC/GV/DDC) Let the difference between two successive ages be $x$. Then the difference between the oldest two and the youngest two ages is $4x = 16$ so $x = 4$. Cai and Denise's ages are 4 apart and add to 34, so they must be 15 and 19,

hence (D).

*Alternative 2*

(PC1/AK) The sum of all 4 ages is $18 + 34 = 52$ so that the average age is $52 \div 4 = 13$. Let the spacing of ages be $2y$ years, so that the ages are $a = 13 - 3y$, $b = 13 - y$, $c = 13 + y$ and $d = 13 + 3y$. Then $c + d = (13 + y) + (13 + 3y) = 26 + 4y$, but also $c + d = 34$, so that $4y = 8$ and $y = 2$. Then $d = 13 + 3 \times 2 = 19$,

hence (D).

**16.** The country of Exponentia uses six-digit telephone numbers. At the moment, this is plenty, since there are only 1000 phone numbers in use. However, increasing population and phone usage means that the number of phone numbers needs to double each year. Approximately how many years will it take for Exponentia to run out of phone numbers?

(A) 5 (B) 10 (C) 20 (D) 30 (E) 40

*Alternative 1*

(MC/AK) The number of phone numbers in use after $n$ years is approximately $1000 \times 2^n$. There are $10^6 = 1\,000\,000$ possible 6-digit phone numbers, or possibly only 900 000 if we assume that numbers starting with 0 are not used. Solving $1000 \times 2^n = 1000000$ gives $2^n = 1000$. Since $2^9 = 512$ and $2^{10} = 1024$, the solution is between $n = 9$ and $n = 10$. Even with the figure of 900 000 phone numbers, the number of years is between 9 and 10. So the closest prediction is that phone numbers will run out after 10 years,

hence (B).

*Alternative 2*

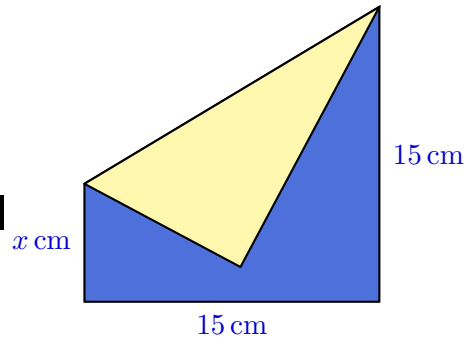
(AK) It is reasonable to assume that most of the 1 million numbers from 000 000 to 999 999 will be available. This table is made by simply following the annual doubling pattern until 1 million (1000 thousand) subscribers is passed:

| Years from now | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Thousands of Subscribers | 1 | 2 | 4 | 8 | 16 | 32 | 64 | 128 | 256 | 512 | 1024 |

So the phone numbers run out in between 9 and 10 years from now,

hence (B).

**17.** A 15 cm × 15 cm square of origami paper is dark blue on top and pale yellow underneath. The top-left corner is folded down so that a crease is made from the top-right corner to a point $x$ cm above the bottom-left corner.

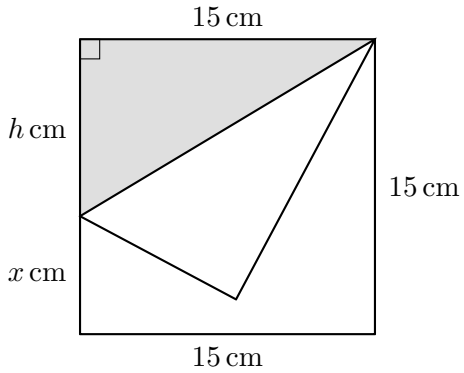



Once folded, the visible regions of yellow and blue paper have equal areas. What is the value of $x$?

(A) 5 (B) $6\frac{2}{3}$ (C) $3\sqrt{3}$ (D) 6 (E) $4\sqrt{2}$

---

Completing the diagram of the square as shown, we have two congruent triangles and a concave pentagon all with equal area. In particular, the shaded triangle with base 15 cm and height $h$ cm has one-third the area of the whole square.




Solving $\frac{1}{2} \times 15 \times h = \frac{1}{3} \times 15^2$ gives $h = 10$, so $x = 15 - h = 5$, hence (A).

**18.** In this equation, the coefficient of $y$ has been hidden, but we know that it is a positive integer, 1 or more.

$$2x + \blacksquare\, y = 25$$

The equation has at least one solution where $x$ and $y$ are positive integers.

How many different values are possible for the hidden coefficient?

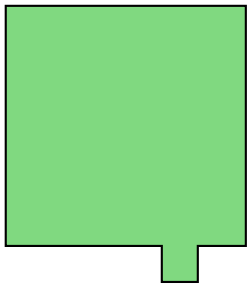
(A) 10 (B) 12 (C) 13 (D) 24 (E) 25

(HG/GV) Let $c$ be the hidden coefficient, and let $(x, y)$ be one of the positive-integer solutions to $2x + cy = 25$ for this value of $c$. Then $2x$ is one of the 12 numbers $2, 4, \ldots, 24$. and $cy$ will be an odd number, one of $23, 21, 19, \ldots, 1$.

So $c$ a divisor of one of the numbers $23, 21, 19, \ldots, 1$, which means that $c$ is odd and $c \leqslant 23$, so $c$ is one of the 12 values $23, 21, 19, \ldots, 1$. For each such value of $c$, there at least one solution, such as $y = 1$, $x = \frac{1}{2}(25 - c)$. So there are exactly 12 possible values of $c$,

hence (B).

**19.** Farmer Smith had a square property that he extended by buying a smaller square of land, creating the property shown.



The new square of land increased the total perimeter of the property by 10%.

By what percentage did the area of the property increase?

(A) 2 (B) 4 (C) 6 (D) 8 (E) 10

*Alternative 1*

(OY/PC) Let the length side of the initial block be $x$ and the length side of the new attachment be $y$. Then, the initial perimeter was $4x$ and the new perimeter is $4x + 2y$. Then, $1.1 \cdot 4x = 4x + 2y$. Hence, $y = 0.2x$. Therefore, the new area is $x^2 + (0.2x)^2 = 1.04x^2$ and the area of the block was increased by 4%,

hence (B).

*Alternative 2*

(AK/GV) Choose a unit so that the large square has side 10 units, with area 100 square units and perimeter 40 units. In the extended property, the perimeter is 44 units.

The additional 4 units of perimeter are purely in the amount of vertical boundary, and is equal to two sides of the small square. So the small square has side 2 units.

Consequently the extended area is $100 + 2^2 = 104$ square units, a 4% increase,

hence (B).

**20.** I hear that dogs age 7 dog years every year. My dog Ruby was born on my ninth birthday. Four years from now, on our birthday, Ruby's age in dog years will be exactly four times my age in normal years. How old am I now?

(A) 10 (B) 12 (C) 13 (D) 15 (E) 17

*Alternative 1*

(ZB/GV) When I am $9 + x$ years old, Ruby will be $7x$ dog-years old. So on our birthday in 4 years time,

$$\begin{aligned} 7x &= 4(9 + x) = 36 + 4x \\ 3x &= 36 \\ x &= 12 \end{aligned}$$

That is, I will be $9 + 12 = 21$ years old then. So I am now 17 years old,

hence (E).

*Alternative 2*

(PC/AK) This table shows our ages on my birthdays

| My age in normal years | 9 | 10 | 11 | 12 | ... |
|---|---|---|---|---|---|
| 4× my age | 36 | 40 | 44 | 48 | ... |
| Ruby's age in dog-years | 0 | 7 | 14 | 21 | ... |
| Difference | 36 | 33 | 30 | 27 | ... |

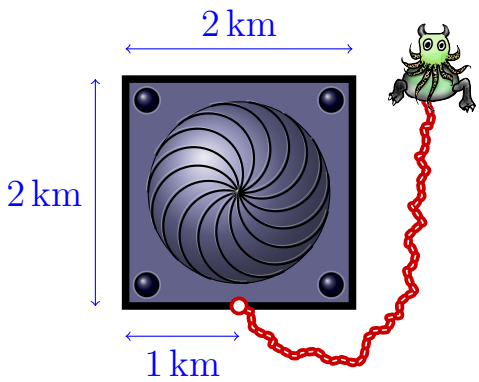
The difference decreases by 3 each year, from the difference between an increase of 4 and an increase of 7. This will decrease to 0 after 12 years, which is my 21st birthday. So I am now 17,

hence (E).

**21.** An ancient beast guards a 2 km × 2 km square building on an otherwise featureless plain.

A 4 km-long unbreakable chain connects the beast to the outside wall of the building, as shown in the diagram. Neither the beast nor the chain can cross into the area occupied by the building.

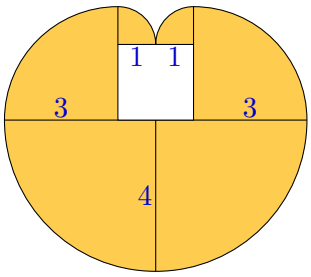
What is the area that the beast can access, in square kilometres?




(A) $9\pi$ (B) $10\pi$ (C) $11\pi$ (D) $12\pi$ (E) $13\pi$

(AK/GV)

Starting with the chain tight and running 4 km south, and heading east, the beast traces quarter-circles of radius 4, 3 and 1 kilometers as the chain wraps around the wall of the building. The western half of the diagram is a mirror image.




The beast can access any point within these quarter-circles, but not the building. The area the beast can access is $2 \times \left(\frac{\pi}{4}1^2 + \frac{\pi}{4}3^2 + \frac{\pi}{4}4^2\right) = \frac{\pi}{2}(1+9+16) = 13\pi\,\text{km}^2$,

hence (E).

**22.** I have four numbers. When I add 3 to the first number, subtract 3 from the second number, multiply the third number by 3 and divide the fourth number by 3, my four answers are all equal.

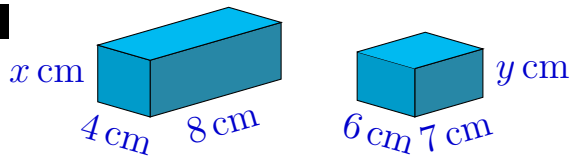
My original 4 numbers added to 32. What is the sum of the largest two of these?

(A) 24 (B) 25 (C) 26 (D) 27 (E) 28

(AK/GV) Let the four numbers be $a, b, c$ and $d$. Then $a+3 = b-3 = 3c = d \div 3$. Writing each in terms of $c$, the four numbers are $a = 3c-3$, $b = 3c+3$, $c$ and $d = 9c$. These add to $16c$, so $16c = 32$ and $c = 2$. Thus $a = 3$, $b = 9$, $c = 2$ and $d = 18$. The largest two add to $9 + 18 = 27$,

hence (D).

**23.** These two rectangular prisms have the same surface area. Both $x$ and $y$ are integers less than 10. What is $x + y$?




(A) 5 (B) 7 (C) 11 (D) 12 (E) 13

---

*Alternative 1*

WA/PC/GV) The surface area of the first prism is $2(32 + 8x + 4x) = 64 + 24x$ and the surface area of the second is $2(42 + 7y + 6y) = 84 + 26y$

So, $64 + 24x = 84 + 26y$ and $x = \dfrac{10 + 13y}{12}$

For $x$ to be an integer, $y$ must be even.

Check

| $y$ | 2 | 4 | 6 | 8 |
|---|---|---|---|---|
| $x$ | 3 | $\frac{62}{12}$ | $\frac{62}{12}$ | $\frac{114}{12}$ |

So, the only possible value for $x + y$ is $2 + 3 = 5$,

hence (A).

*Alternative 2*

Equating surface areas,

$$2 \times (32 + 4x + 8x) = 2 \times (42 + 6y + 7y)$$
$$12x = 10 + 13y$$

One integer solution is $x = y = -10$. However $x$ and $y$ should both be positive. Other integer solutions can be found by increasing $x$ by 13 and $y$ by 12, since then both sides of the equation increase by $12 \times 13 = 156$. This gives the solution $x = 3$, $y = 2$.

It seems that this is the only solution. To confirm, compare the solution $x = 3$, $y = 2$ with some other solution $x = a$, $y = b$. Then

$$12a = 10 + 13b$$
$$12 \times 3 = 10 + 13 \times 2$$
$$12(a - 3) = 13(b - 2)$$

So this difference must be a multiple of $\text{lcm}(12, 13) = 156$. However, $a - 3$ and $b - 2$ are both between $-3$ and 7, so that the only possible multiple of 156 is 0.

Thus $x = 3$, $y = 2$ is the only solution with $x$ and $y$ positive integers less than 10, and then $x + y = 5$,

hence (A).

**24.** Sometimes a three-digit number is an exact multiple of its digit sum. For instance, the digit sum of 102 is $1 + 0 + 2 = 3$ and $102 = 3 \times 34$. If a three-digit number is $k$ times the sum of its digits, what is the smallest possible integer value of $k$?

(A) 9 (B) 10 (C) 11 (D) 12 (E) 13

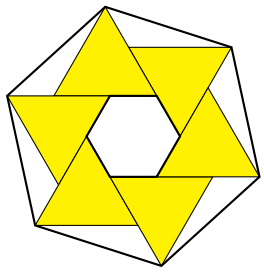
(MC/AK) Since $100a + 10b + c = k(a + b + c)$, the expression $X = a(100 - k) + b(10 - k) + c(1 - k)$ has $X = 0$. Since $X$ requires both positive and negative terms, $1 < k < 100$.

However, if $1 < k \leqslant 10$, then $X \geqslant 1(100 - 10) + (10 - 10) - 9(10 - 1) = 9 > 0$, contradicting $X = 0$, so that $k \leqslant 10$ is not possible. Consequently $11 \leqslant k < 100$.

Trying $k = 11$, $X = 89a - b - 10c$ so that $X = 0$ when $b = 9$, $c = 8$. This corresponds to the solution $198 = 11(1 + 9 + 8)$, so that $k = 11$ is the smallest possible value,

hence (C).

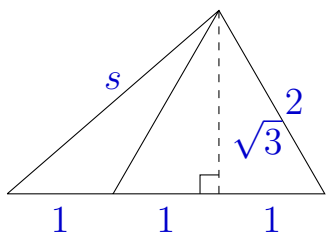
**25.** Six identical equilateral triangles of side length 2 are drawn outside a regular hexagon of side length 1, defining a larger hexagon as shown. What is the ratio of the area of the larger hexagon to the area of the smaller hexagon?



(A) 5 : 1 (B) 6 : 1 (C) 7 : 1 (D) 8 : 1 (E) 9 : 1

*Alternative 1*

(GV) Consider the following section of the diagram, including one of the equilateral triangles and one of the larger hexagon's sides, from the top left of the hexagon.

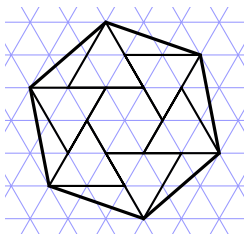



The dotted line is a median of the equilateral triangle, so divides it into two right-angled triangles of sides 1, 2 and $\sqrt{3}$. By Pythagoras' theorem, $s^2 = 2^2 + (\sqrt{3})^2 = 7$, so that $s$, the side of the large hexagon, is $\sqrt{7}$. Consequently the ratio of the sides of the two hexagons is $s : 1 = \sqrt{7} : 1$. Since the two hexagons are similar, the ratio of the areas is $s^2 : 1 = 7 : 1$,

hence (C).

*Alternative 2*

A grid of equilateral triangles of side 1 can be aligned with the hexagon and the equilateral triangles



In units from the grid, the small hexagon has area 6 and the large hexagon has area $6 + 6 \times 4 + 6 \times (4 \div 2) = 42$. Then the ratio of areas is $42 : 6 = 7 : 1$,

hence (C).

**26.** Seána was arranging her collection of postage stamps into groups when a cat jumped onto them and scattered the stamps. All she can remember is that when she put them into groups of 2, 3, 4, 5 or 6 she always had 1 stamp left over. When she placed them into groups of 7 there were none left over.

What is the minimum number of stamps Seána could have had in her collection?

---

*Alternative 1*

If Seána has $N$ stamps, then $N-1$ is a multiple of $\text{lcm}(2,3,4,5,6) = 60$. Also $N$ is a multiple of 7, the answer is the smallest number of the form $60k+1$ that is a multiple of 7. Since $60 \div 7 = 8\frac{4}{7}$, $(60k+1) \div 7 = 8k + \frac{1}{7}(4k+1)$, so that $4k+1$ is a multiple of 7. We check values of $k$:

| $k$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $4k+1$ | 1 | 5 | 9 | 13 | 17 | 21 |

Then $k=5$ is the smallest value for which $4k+1$ is a multiple of 7, and for which $60k+1 = 301$ is a multiple of 7. So there are 301 stamps in Seána's collection,

hence (301).

*Alternative 2*

(GV/AK) Since there is one stamp over when dividing by 2 and by 5, the last digit in $N$ (the number of stamps) must be 1.

Also $N$ is a multiple of 7, so it will be $N = 7M$ where $M$ has last digit 3.

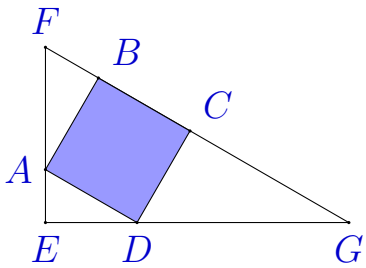
Now check $M = 3, 13, 23, \ldots$ for the first case where all other remainders are 1:

| $M$ | 3 | 13 | 23 | 33 | 43 |
|---|---|---|---|---|---|
| $N$ | 21 | 91 | 161 | 231 | 301 |
| Remainder in $N \div 3$ | 0 | 1 | 2 | 0 | 1 |
| Remainder in $N \div 4$ | 1 | 3 | 1 | 3 | 1 |
| Remainder in $N \div 6$ | 3 | 1 | 5 | 3 | 1 |

So the smallest possibility is to have 301 stamps,

hence (301).

**27.** A square $ABCD$ is inscribed in a right-angled triangle $EFG$ as shown.
The length of $EG$ is 4 units and the length of $EF$ is 3 units.
As a fraction in simplest form, the side-length of the square is $\frac{a}{b}$. What is the value of $a+b$?

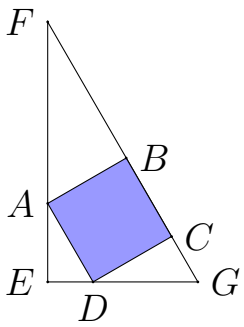



*Alternative 1*

(KH/PC)

$ABCD$ is the square inside the triangle $EFG$. Let $AB = x$.
Triangle $EFG$ is right-angled with $EG = 3$ and $EF = 4$. Hence $FG = 5$. Clearly triangles $EFG$, $BFA$ and $CDG$ are similar. So within $BFA$, as $AB = x$ then $BF = 4x/3$. And within $CDG$, as $CD = x$ then $CG = 3x/4$.
Hence $FG = FB + BC + CG = 4x/3 + x + 3x/4 = 37x/12$. But $FG = 5$. Hence $x = \frac{60}{37}$.
Since this is in simplest terms, $a = 60$, $b = 37$ and $a + b = 97$,

hence (97).

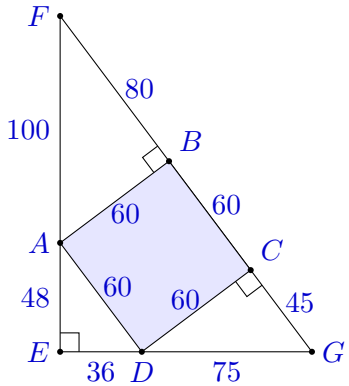



*Alternative 2*

(AK)

By Pythagoras' theorem, $FG = 5$ units. The three unshaded right-angled triangles are similar to $\triangle EFG$ with ratios $3:4:5$, and with one side on the square.
Define a temporarily unit 'jot' so that the square has side $\text{lcm}(3,4,5) = 60$ jots. The other sides shown are calculated using the $3:4:5$ ratios.
Then $EG = 111$ jots, $EF = 148$ jots and $FG = 185$ jots, which is 37 times the lengths in the original units. Hence 1 jot is $\frac{1}{37}$ units and the square has side $\frac{a}{b} = \frac{60}{37}$ units. Since 60 and 37 have no common factor, $a = 60$, $b = 37$ and $a + b = 97$,

hence (97).



**28.** The elves have to choose who will go the annual magic conference. They sit in a circle and the chief elf Elvin starts counting round the circle, starting with himself. Every second elf counted drops out of the circle and the counting continues until Elvin drops out. All those left in the circle go to the conference.

This year, there are 1000 elves in the circle. How many will go to the conference?

---

IC/AK/DDC) The count goes around the circle several times, with odd-numbered Elves staying and even-numbered Elves dropping out. The stay/drop pattern is not affected if an even number is subtracted at any point, so we can assume whenever Elvin is due to count an odd number, he restarts at "1". The count goes around the circle 4 times:

| Elves before cycle | Count | Elves after cycle |
|---|---|---|
| 1000 | 1, ~~2~~, 3, ~~4~~, . . . , ~~998~~, 999, ~~1000~~ | 500 |
| 500 | 1, ~~2~~, 3, ~~4~~, . . . , ~~498~~, 499, ~~500~~ | 250 |
| 250 | 1, ~~2~~, 3, ~~4~~, . . . , ~~248~~, 249, ~~250~~ | 125 |
| 125 | 1, ~~2~~, 3, ~~4~~, . . . , 123, ~~124~~, 125 | 63 |

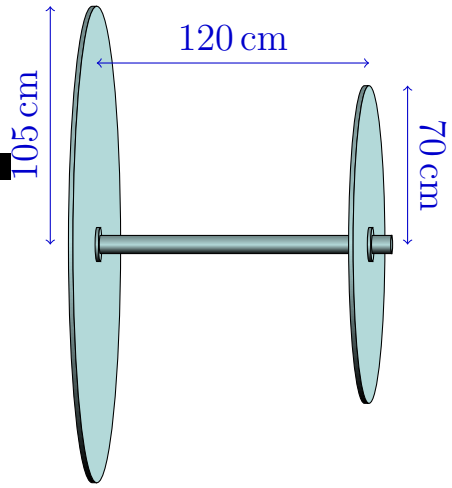
At this point the pattern changes, since Elvin is due to count an even number. So he drops out immediately, leaving 62 elves,

hence (62).

**29.** Two wheels are fixed to an axle as shown.

Due to their different sizes, the two wheels trace two concentric circles when [redacted] rolled on level ground.

In centimetres, what is the radius of the circle traced on the ground by the larger wheel?




---

*Alternative 1*

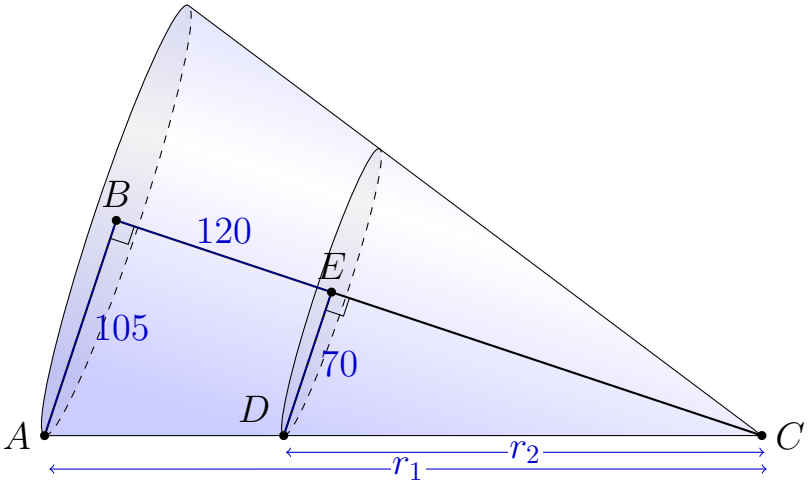
(AK) For any distance rolled, the larger wheel travels $\frac{105}{70} = \frac{3}{2}$ as far as the smaller wheel. So the circumference of the larger traced circle is $\frac{3}{2}$ that of the smaller traced circle, which means the radii are also in the same ratio. Consequently the distance between the two traced circles is half the radius of the smaller circle, and also one-third the radius of the larger circle.

The difference between these two radii is the distance between the two wheel rims, which is $\sqrt{(105-70)^2+120^2} = \sqrt{35^2+120^2} = 5\sqrt{7^2+24^2} =$ [redacted] $5 \times \sqrt{625} = 5 \times 25 = 125\,\text{cm}$. Consequently the larger traced circle has radius $375\,\text{cm}$,

hence (375).

*Alternative 2*

(PC/AK) The pair of wheels can be imagined as part of a cone, and we consider a points $A, B, C, D, E$ in a vertical plane as shown below. The rolling of the wheels is equivalent to rolling the cone with its apex $C$ fixed. Then the radii of the two circles traced by the wheels are $r_1 = AC$ and $r_2 = CD$.



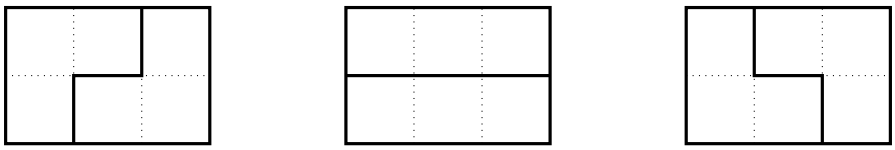

Then $\triangle ABC$ and $\triangle DEC$ are similar with $DE = \frac{2}{3}AB$ so that $EC = \frac{2}{3}BC$. Then $BE = 120 = \frac{1}{3}BC$ so that $BC = 360$ and $EC = 240$.

Rather than using Pythagoras' theorem directly to work out $AC$, observe the ratio $EC : DE = 24 : 7$ in $\triangle DEC$. Then $24^2 + 7^2 = 576 + 49 = 625 = 25^2$ so it follows that both triangles are in the proportions of the Pythagorean triple $(7, 24, 25)$.

With these proportions, $\triangle ABC$ has sides $AB = 15 \times 7 = 105$, $BC = 15 \times 24 = 360$ and $AC = r_1 = 15 \times 25 = 375$,

hence (375).

**30.** A tromino is a shape made from three squares traced on gridlines. A $2 \times 3$ grid can be tiled by trominoes in exactly three ways, as shown.



We count two tilings that are reflections of each other as different. Similarly, two tilings that are rotations of each other are counted as different.

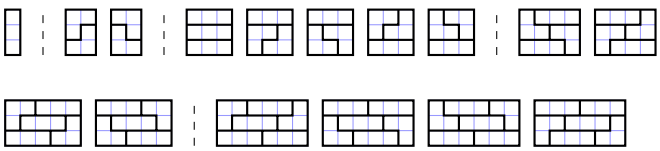
In how many different ways can a $3 \times 6$ grid be tiled by trominoes?

---

*Alternative 1*

(6x3) Orient the rectangle to be 6 wide and 3 high. Each tiling will have a pattern of *breaks*, which we define as being vertical lines that extend the full height of the rectangle. Between breaks is an *unbroken* tiling of a $k \times 3$ rectangle.

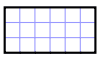
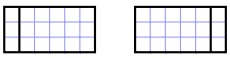
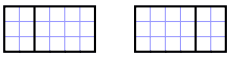
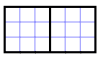
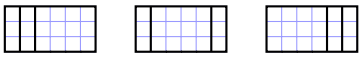
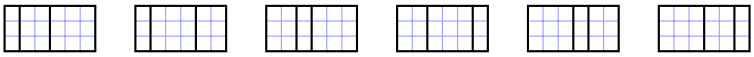
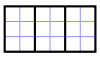
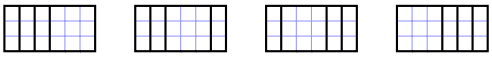
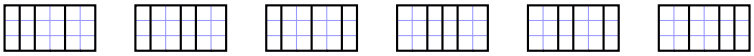
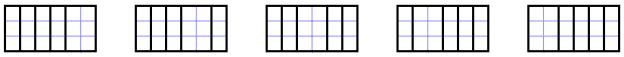
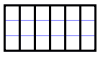
By carefully working through possibilities, we can find all possible unbroken tilings of widths 1 to 6:



| Width | Tilings |
|---|---|
| 1 | 1 |
| 2 | 2 |
| 3 | 5 |
| 4 | 2 |
| 5 | 2 |
| 6 | 4 |

Now consider the "templates" of vertical breaks. There are $2^5 = 32$ possible templates, corresponding to whether there is or is not a break at each position. We organise these in the table below by the number "blocks" of each width, and use multiplication to find the number of ways to fill each template.

For instance for the third row, there are two templates corresponding to the two ways of combining blocks of widths 2 and 4. For each of these templates the $2 \times 3$ block can be tiled in 2 unbroken ways and the $4 \times 3$ block can be tiled in 2 unbroken ways. So there are $2 \times 2 = 4$ tilings with each template, and $2 \times 4 = 8$ tilings for that row.

| Vertical break template(s) | | Tilings per template | Tilings |
|---|---|---|---|
|  | 1 | 4 | 4 |
|  | 2 | 2 | 4 |
|  | 2 | $2 \times 2$ | 8 |
|  | 1 | $5 \times 5$ | 25 |
|  | 3 | 2 | 6 |
|  | 6 | $2 \times 5$ | 60 |
|  | 1 | $2 \times 2 \times 2$ | 8 |
|  | 4 | 5 | 20 |
|  | 6 | $2 \times 2$ | 24 |
|  | 5 | 2 | 10 |
|  | 1 | 1 | 1 |
| | | | 170 |

The calculation in this table gives 170 possible tilings,

hence (170).

*Alternative 2*

(6x3 alt) Similar to above, but work out number of tilings of $n \times 3$ by recurrence relation where a 'broken' tiling has the last block removed. So we have $U_n$, the number of unbroken tilings from the previous solution, then $B_n$ the number of broken tilings, and $T_n = U_n + B_n$.

| $n$ | $U_n$ | $B_n$ | | $T_n$ |
|---|---|---|---|---|
| 1 | 1 | | 0 | 1 |
| 2 | 2 | $T_1U_1$ | $= 1$ | 3 |
| 3 | 5 | $T_1U_2 + T_2U_1$ | $= 2 + 3 = 5$ | 10 |
| 4 | 2 | $T_1U_3 + T_2U_2 + T_3U_1$ | $= 5 + 6 + 10 = 21$ | 23 |
| 5 | 2 | $T_1U_4 + T_2U_3 + T_3U_2 + T_4U_1$ | $= 2 + 15 + 20 + 23 = 60$ | 62 |
| 6 | 4 | $T_1U_5 + T_2U_4 + T_3U_3 + T_4U_2 + T_5U_1$ | $= 2 + 6 + 50 + 46 + 62 = 166$ | 170 |

# 3 Senior solutions

**1.** What is the value of $2023 - 3202$?

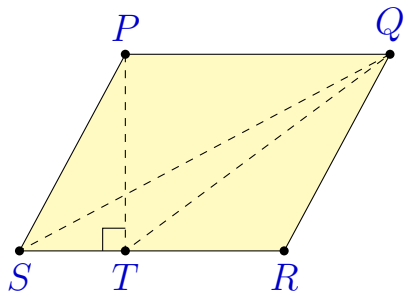
(A) $-1221$ (B) $-1179$ (C) $1179$ (D) $1221$ (E) $5225$

(PC) $2023 - 3202 = -1179$

---

**2.** A parallelogram $PQRS$ has an area of $60\,\text{cm}^2$ and side $PQ$ of length $10\,\text{cm}$. Which length is $6\,\text{cm}$?

(A) $RQ$ (B) $RS$ (C) $QT$ (D) $PT$ (E) $QS$




(AK) The parallelogram's area is $A = bh$ where $b$ the base and $h$ the height perpendicular to that base. With base $b = PQ = 10\,\text{cm}$ and area $A = 60\,\text{cm}^2$ we must have $h = 6$. Of the given segments, $PT$ is perpendicular $PQ$ so $PT = 6\,\text{cm}$. All other given segments are longer, hence (D).

---

**3.** Which one of these is equal to $57 \times 953$?

(A) 321 (B) 4321 (C) 54 321 (D) 654 321 (E) 7 654 321

Rounding up, $57 \times 953 < 60 \times 1000 = 60\,000$, which suggests (C). Rounding down, $57 \times 953 > 50 \times 900 = 45\,000$, so that only (C) is possible. Evaluating precisely confirms that $57 \times 953 = 54321$, hence (C).

---

**4.** What is the difference between $2^5$ and $5^2$?

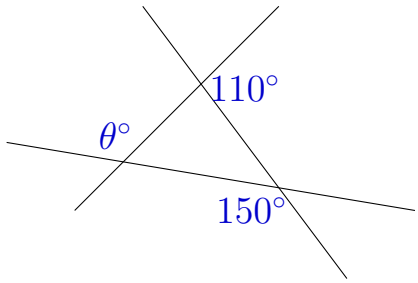
(A) 0 (B) 1 (C) 3 (D) 5 (E) 7

The difference between $2^5 = 32$ and $5^2 = 25$ is $32 - 25 = 7$, hence (E).

---

**5.** What is the value of the angle $\theta°$ in the diagram?

(A) 100° (B) 110° (C) 120° (D) 130° (E) 140°



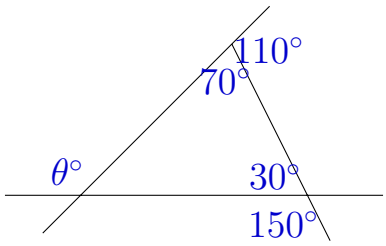

*Alternative 1*

The three exterior angles will add to 360° so $\theta = 360 - (110 + 150) = 100°$,

hence (A).

*Alternative 2*

(GV) Using supplementary angles, two interior angles can be found




Then $\theta = 70 + 30 = 100$,

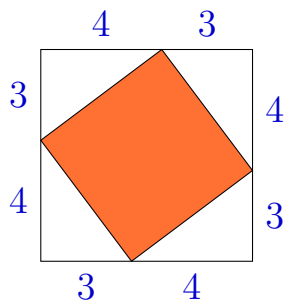
hence (A).

---

**6.** The shaded square is inscribed in the larger square as shown.

What is the ratio of shaded to unshaded area in the diagram?

(A) 5 : 4 (B) 25 : 24 (C) 3 : 2 (D) 7 : 4 (E) 12 : 7




The side of the shaded square is $\sqrt{3^2 + 4^2} = 5$ so its area is $5^2 = 25$.
There are 4 unshaded triangles, with total area $4 \times \frac{1}{2} \times 3 \times 4 = 24$.
So the ratio of shaded to unshaded is 25 : 24,

hence (B).

---

**7.** Jemmy multiplies together all the integers from 1 to 18. What are the last three digits of the result?

(A) 000 (B) 020 (C) 200 (D) 080 (E) 800

The numbers being multiplied include 4, 5, 10 and 15. Multiplying these gives 3000. Multiplying this by the remaining numbers in the list will always leave three zeros at the end,

hence (A).

---

**8.** A fuel tank is 40% empty. Then 40 litres of fuel is removed. The tank is now 40% full. How many litres are in a full tank?

(A) 40 (B) 100 (C) 160 (D) 200 (E) 400

*Alternative 1*

(WA2/GV) Let the full volume be $v$. Then

$$\begin{aligned} \frac{60}{100}v &= \frac{40}{100}v + 40 \\ 60v &= 40v + 4000 \\ 20v &= 4000 \\ v &= 200, \end{aligned}$$

hence (D).

*Alternative 2*

(AK/PC) The tank is initially 40% empty, which is 60% full, and then it becomes 40% full. So the 40 L removed is 20% of the tank. Then $5 \times 40 = 200$ L is 100% of the tank,

hence (D).

---

**9.** The volume $V$ of a sphere of radius $r$ is given by $V = \frac{4}{3}\pi r^3$.

For a sphere of volume $V = 100\,\text{cm}^3$, which of the following is closest to the radius?

(A) 2.9 cm (B) 3.5 cm (C) 5 cm (D) 5.8 cm (E) 10 cm

---

Solving for $r$,

$$\begin{aligned} 100 &= \frac{4}{3}\pi r^3 \\ \frac{75}{\pi} &= r^3 \\ r &= \sqrt[3]{\frac{75}{\pi}} \approx \sqrt[3]{\frac{75}{3}} = \sqrt[3]{25} \approx \sqrt[3]{27} = 3 \end{aligned}$$

This suggests the closest answer is (A).

More precisely, we need to confirm that $r < 3.2$ for (A) to be closer than (B). So $r^3 = \frac{75}{\pi} < \frac{81}{3} = 27 = 3^3$ so that $r < 3$. Out of the 5 given values, the closest must be 2.9,

hence (A).

**10.** If I add 3 consecutive odd integers, I get a total of $9m+3$. The largest of these 3 integers is

(A) $3m-3$ (B) $3m-1$ (C) $3m$ (D) $3m+1$ (E) $3m+3$

*Alternative 1*

The average of the three numbers is $(9m+3)\div 3 = 3m+1$. The three numbers are evenly spaced 2 apart, so the middle one is the average of all three. Then they are $3m-1, 3m+1$, and $3m+3$, with largest $3m+3$,

hence (E).

*Alternative 2*

(PC) Let the largest of the three be $x$, so that all three are $x-4$, $x-2$ and $x$. Then

$$\begin{aligned}(x-4)+(x-2)+x &= 9m+3\\ 3x-6 &= 9m+3\\ 3x &= 9m+9 = 9(m+1)\\ x &= 3(m+1) = 3m+3\end{aligned}$$

hence (E).

---

**11.** The value of $(\sqrt{24}+\sqrt{54})^2$ is

(A) 140 (B) 150 (C) 160 (D) 170 (E) 180

*Alternative 1*

(author) Noticing that $24 = 2^3\times 3$ and $54 = 2\times 3^3$, we see that

$$\sqrt{24}\times\sqrt{54} \;=\; \sqrt{2^4\times 3^4} \;=\; 2^2\times 3^2 \;=\; 36\,.$$

Therefore

$$(\sqrt{24}+\sqrt{54})^2 \;=\; 24 \;+\; (2\times 36) \;+\; 54 = \;150$$

hence (B).

*Alternative 2*

(AK/PC/GV) Simplifying each radical, $(\sqrt{24}+\sqrt{54})^2 = (2\sqrt{6}+3\sqrt{6})^2 = (5\sqrt{6})^2 = 25\times 6 = 150$,

hence (B).

---

**12.** In a group of 6 people there are 3 pairs of twins. How many 3-member committees can be chosen that do not contain any pair of twins?

(A) 0 (B) 8 (C) 12 (D) 24 (E) 48

(AH/GV) Let the twins be $\{A, B\}$, $\{C, D\}$, $\{E, F\}$. Every choice of committee must choose 1 from each pair of twins, so there are $2\times2\times2 = 8$ possible committees. These can be listed: $\{ACE, ACF, ADE, ADF, BCE, BCF, BDE, B$

hence (B).

---

**13.** Assuming $a > b > 0$, the expression $\dfrac{a^{-1} - b^{-1}}{a^{-2} - b^{-2}}$ can be written as

(A) $\frac{b-a}{ab}$ (B) $\frac{b+a}{ab}$ (C) $\frac{ab}{b+a}$ (D) $\frac{a^2-b^2}{a-b}$ (E) $a - b$

*Alternative 1*

(JB/PC)

$$\begin{aligned}\frac{\frac{1}{a}-\frac{1}{b}}{\frac{1}{a^2}-\frac{1}{b^2}} &= \frac{b-a}{ab} \div \frac{b^2-a^2}{a^2b^2} \\ &= \frac{(b-a)}{ab} \times \frac{a^2b^2}{(b-a)(b+a)} \\ &= \frac{ab}{a+b}\end{aligned}$$

hence (C).

*Alternative 2*

(AK2) The denominator is a difference of squares: $a^{-2} - b^{-2} = (a^{-1} - b^{-1})(a^{-1} + b^{-1})$ so the expression cancels to $X = \dfrac{1}{a^{-1}+b^{-1}}$. Then $a^{-1} + b^{-1} = \frac{1}{a} + \frac{1}{b} = \frac{b+a}{ab}$ and so $X = \frac{ab}{b+a}$,

hence (C).

---

**14.** A bag contains red and yellow balls such that the ratio red : yellow is $5:7$.

Then 10 balls of each colour are removed and the ratio changes to $5:8$.

How many balls were originally in the bag?

(A) 48 (B) 60 (C) 72 (D) 84 (E) 96

---

*Alternative 1*

(KH) Suppose there were $x$ red balls and $y$ yellow balls at the start, then $x:y=5:7$ and so $7x=5y$.

After 10 each is removed then $(x-10):(y-10)=5:8$

Hence $8(x-10)=5(y-10)$ and so $8x-80=5y-50$

But $5y=7x$, so $8x-80=7x-50$

Thus $x=30$ and $y=42$. Checking, $30:42=5:7$ and $20:32=5:8$. So there were $30+42=72$ balls originally,

hence (C).

*Alternative 2*

(GV) At first there were $5a$ red balls and $7a$ yellow balls. Then there were $5a-10$ red and $7a-10$ yellow, in the ratio $5:8$. Solving,

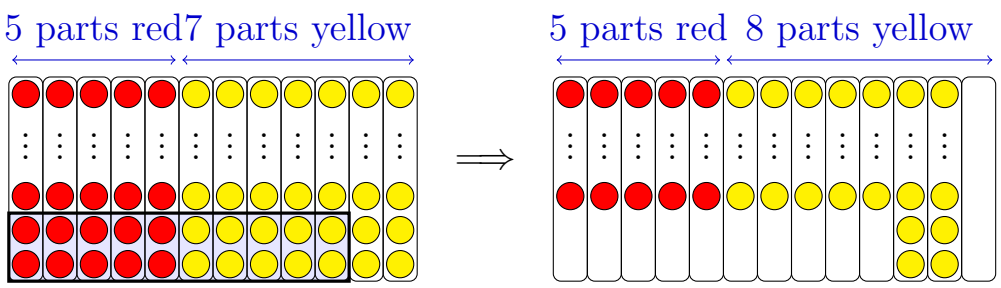
$$\begin{aligned}(5a-10):(7a-10)&=5:8\\ 8(5a-10)&=5(7a-10)\\ 40a-35a&=80-50\\ a&=6\end{aligned}$$

Initially there were $5a+7a=12a=72$ balls,

hence (C).

*Alternative 3*

(AK) The balls from the bag can be arranged in 12 equal columns as shown on the left:

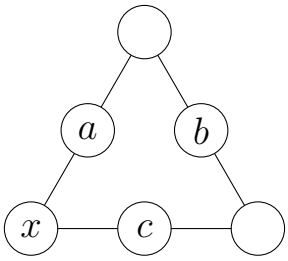


After the 20 balls are removed , 4 yellow balls remain in the bottom two rows. The new ratio of $5:8$ needs an extra yellow column, equal to all the other columns. So the 4 extra balls must match this last column, and all final columns have 4 balls. Then the initial columns had 6 balls each with a total of $12 \times 6 = 72$ balls,

hence (C).

---

**15.** In the solution to this number puzzle, whenever there are three numbers in a straight line, the middle number is the sum of the other two.




What is the value of $x$?

(A) $\frac{1}{2}(a+b+c)$ (B) $\frac{1}{2}(b-a-c)$ (C) $\frac{1}{2}(a+b-c)$

(D) $\frac{1}{2}(a-b+c)$ (E) $\frac{1}{2}(b+c-a)$

---

*Alternative 1*

(WA/GV) Let the other two end numbers be $y$ and $z$.

Then

$$\begin{array}{rrcll}
 & x+y & = & a & \text{A} \\
 & y+z & = & b & \text{B} \\
 & x+z & = & c & \text{C} \\
\text{C} - \text{B} & x-y & = & c-b & \text{D} \\
\text{A} + \text{D} & 2x & = & a-b+c & \\
 & x & = & \frac{1}{2}(a-b+c) &
\end{array}$$

hence (D).

*Alternative 2*

(AK) The three edge numbers $a+b+c$ add to twice the sum of the corner numbers. Also, the sum of the 3 corner numbers is $x+b$. So we solve

$$a+b+c = 2(x+b) \implies 2x = a-b+c \implies x = \tfrac{1}{2}(a-b+c)$$

hence (D).

---

**16.** If $f(x) = 5 + x$ and $g(x) = 7 - x$, then $f(g(x)) - g(f(x))$ equals

(A) $10 - x$ (B) $7$ (C) $x + 2$ (D) $10$ (E) $2x - 2$

$$\begin{aligned} f(g(x)) - g(f(x)) = f(5+x) - g(7-x) &= 5 + (7-x) - (7-(5+x)) \\ &= 12 - x - 2 + x \\ &= 10 \end{aligned}$$

---

**17.** The hypotenuse of a right-angled triangle has length 6 cm. The perimeter of the triangle is 14 cm.

What is the area of the triangle in square centimetres?

(A) 7 (B) 12 (C) 14 (D) 21 (E) 24

*Alternative 1*

(AH/PC) Suppose the sides are $a, b, 6$. By Pythagoras' theorem, $a^2 + b^2 = 6^2 = 36$. From the perimeter, $a + b + 6 = 14$ so that $a + b = 8$. Then $64 = (a+b)^2 = a^2 + 2ab + b^2 = 36 + 2ab$ so that $ab = \frac{1}{2}(64 - 36) = 28$. Then the area of the triangle is $\frac{1}{2}ab = 14\,\text{cm}^2$,

hence (A).

*Alternative 2*

(AK) Let the sides of the triangle be $x, y$ and 6 so that $y = 14 - 6 - x = 8 - x$. By Pythagoras' theorem,

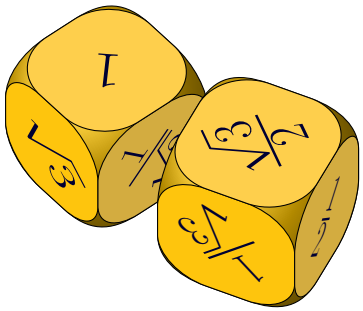
$$\begin{aligned} x^2 + (8-x)^2 &= 36 \\ 2x^2 - 16x + 64 - 36 &= 0 \\ x^2 - 8x + 14 &= 0 \\ x &= \frac{8 \pm \sqrt{64-56}}{2} = 4 \pm \sqrt{2} \text{ (quadratic formula)} \\ y &= 8 - x = 4 \mp \sqrt{2} \\ \frac{xy}{2} &= \frac{1}{2}(4+\sqrt{2})(4-\sqrt{2}) = \frac{16-2}{2} = 7 \end{aligned}$$

hence (A).

---

**18.** I have two identical dice, each with faces

$$1,\ \frac{1}{2},\ \sqrt{3},\ \frac{1}{\sqrt{3}},\ \frac{1}{\sqrt{2}} \quad \text{and} \quad \frac{\sqrt{3}}{2}.$$




I roll both dice and multiply the two numbers rolled, then simplify my answer. What is the probability that this product is rational?

(A) $\frac{1}{9}$ (B) $\frac{1}{6}$ (C) $\frac{1}{2}$ (D) $\frac{5}{18}$ (E) $\frac{7}{18}$

(HG/PC/GV) The sample space has 36 equally likely rolls, as shown in this table. The cases where the product is rational are ticked.

| | $1$ | $\frac{1}{2}$ | $\sqrt{3}$ | $\frac{1}{\sqrt{3}}$ | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ |
|---|---|---|---|---|---|---|
| $1$ | ✓ | ✓ | | | | |
| $\frac{1}{2}$ | ✓ | ✓ | | | | |
| $\sqrt{3}$ | | | ✓ | ✓ | | ✓ |
| $\frac{1}{\sqrt{3}}$ | | | ✓ | ✓ | | ✓ |
| $\frac{1}{\sqrt{2}}$ | | | | | ✓ | |
| $\frac{\sqrt{3}}{2}$ | | | ✓ | ✓ | | ✓ |

Then 14 out of 36 are ticked, so the probability is $\frac{14}{36} = \frac{7}{18}$, hence (E).

**19.** How many distinct pairs $(x, y)$ satisfy $x^2 + y^2 < 50$ if $x$ and $y$ are both positive integers with $x > y$?

(A) 15 (B) 13 (C) 11 (D) 9 (E) 8

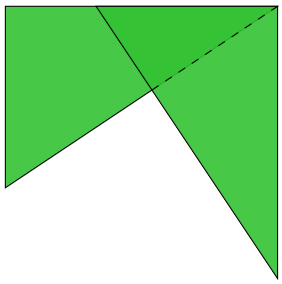
(HG) We have $1 \leqslant y^2 \leqslant x^2 < 50$ where $x^2$ and $y^2$ are chosen from $\{1, 4, 9, 16, 25, 36, 49\}$. Possibilities are

$$\begin{array}{lllll} 1+4, & 1+9, & 1+16, & 1+25, & 1+36, \\ & 4+9, & 4+16, & 4+25, & 4+36, \\ & & 9+16, & 9+25, & 9+36, \\ & & & 16+25. & \end{array}$$

So there are 13 solutions,

hence (B).

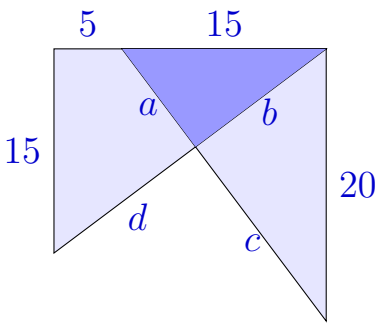
**20.** A 15 cm × 20 cm rectangle is cut into two triangles. One triangle is rotated and placed on top of the other triangle as shown to form a concave pentagon.



What is the perimeter of the pentagon?

(A) 80 cm (B) 84 cm (C) 93 cm (D) 96 cm (E) 105 cm

Due to Pythagoras' theorem, the hypotenuse of each right-angled triangle is 25, and so each is a right-angled triangle with sides in the ratio 3 : 4 : 5.




The smaller triangle formed by the overlap of the two pieces (darker shading) has two angles from the paper pieces, so is similar to these. In particular it has sides in the ratio 3 : 4 : 5 where the hypotenuse is 15. So its other two sides are $a = 9$ and $b = 12$ and then $c = 25 - 9 = 16$ amd $d = 25 - 12 = 13$. The pentagon has perimeter $15 + 20 + 20 + 16 + 13 = 84$,

hence (B).

**21.** I have four numbers. When I add 3 to the first number, subtract 3 from the second number, multiply the third number by 3 and divide the fourth number by 3, my four answers are all equal.

My original 4 numbers added to 32. What is the sum of the largest two of these?

(A) 24 (B) 25 (C) 26 (D) 27 (E) 28

(AK/GV) Let the four numbers be $a, b, c$ and $d$. Then $a+3 = b-3 = 3c = d \div 3$. Writing each in terms of $c$, the four numbers are $a = 3c-3$, $b = 3c+3$, $c$ and $d = 9c$. These add to $16c$, so $16c = 32$ and $c = 2$. Thus $a = 3$, $b = 9$, $c = 2$ and $d = 18$. The largest two add to $9 + 18 = 27$,

hence (D).

**22.** Antonio walked 11.5 km to his cousin Maria's house.

At first he walked uphill, then along a flat part of the road and the final part was downhill. The trip took 2 hours 54 minutes. The next day his walk back home took 3 hours 6 minutes. Antonio walks uphill at a constant speed of 3 km/h, on the flat at 4 km/h and downhill at 5 km/h.

What is the length, in kilometres, of the flat part of the road?

(A) 4 (B) 4.5 (C) 5 (D) 5.5 (E) 6

---

*Alternative 1*

(OY/PC/GV) Let the uphill part of the road be $x$ km, the flat part through the valley $y$ km and the downhill part $z$ km. Then, we get the following system of 3 equations

$$\begin{aligned} \frac{x}{3}+\frac{y}{4}+\frac{z}{5} &= 2.9 \\ \frac{x}{5}+\frac{y}{4}+\frac{z}{3} &= 3.1 \\ x+y+z &= 11.5. \end{aligned}$$

By adding the first two equations we get

$$\frac{8}{15}(x+z)+\frac{y}{2}=6.$$

From the third equation we get $x+z=11.5-y$ and substitute it into the above equation, which yields

$$\frac{8}{15}\left(\frac{23}{2}-y\right)+\frac{y}{2}=6.$$

and we come to the solution $y=4$,

hence (A).

*Alternative 2*

(AK) The three speeds are 12, 15 and 20 minute per kilometre. Consider the round trip of 6 hours = 360 minutes.

Each 1 km of hilly road takes $12+20=32$ minutes to walk both ways. Each 1 km of flat road takes $15+15=30$ minutes to walk both ways. So relative to a flat road, each 1 km of hill adds 2 minutes to the round trip.

If the 11.5 km was all flat, the round trip would take $11.5 \times 30 = 345$ minutes. It took 15 minutes longer, so there must be 7.5 km of hills. Therefore there is 4 km of flat road,

hence (A).

---

**23.** I think of two positive integers and call their sum $S$ and their product $P$. Which one of the following could **not** be the value of $S + P$?

(A) 84 (B) 86 (C) 88 (D) 90 (E) 92

(ND/AK) If the two numbers are $A$ and $B$, then the value of $S + P$ is

$$A + B + AB = (A + 1)(B + 1) - 1.$$

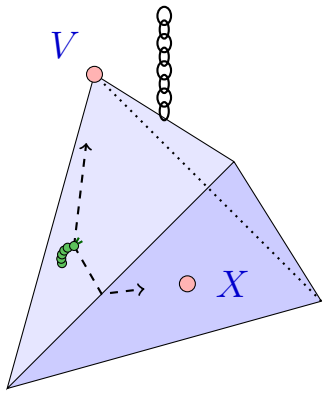
Since $A$ and $B$ are positive integers, this means that $(A + 1)(B + 1)$ cannot be a prime number. So $S + P$ cannot be equal to 1 less than a prime number.

Conversely if $N$ is one less than a composite number, say $N = xy - 1$ where $x \geqslant y \geqslant 2$, then $A = x-1$ and $B = y-1$ have sum $S = x+y-2$ and product $P = xy - x - y + 1$ so that $S + P = N$. Looking at the numbers to choose from $85, 87, 91$ and $93$ are composite whereas 89 is prime, so 88 is the only number that cannot be $S + P$,

hence (C).

---

**24.** A solid regular tetrahedron has 4 faces, each an equilateral triangle. It is suspended in the entomology laboratory. There are two food sources on the tetrahedron, one at a vertex $V$ and the other at $X$, the centre of the opposite face.




When a geodesic grub is placed anywhere upon the tetrahedron, it instinctively crawls along the shortest possible path over the surface to the closest food source.

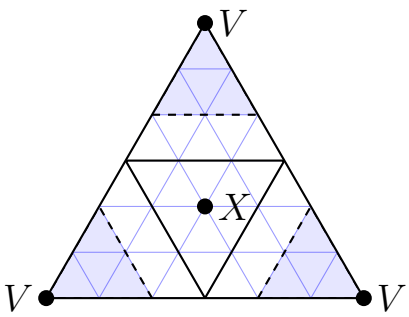
What fraction of the surface area is closer to $V$ than to $X$, in terms of paths along the surface?

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$ (E) $\frac{3}{4}$

---

Open the tetrahedron up to the net shown, where there are 3 copies of food source $V$. This doesn't change the lengths of paths the grub takes, since a path crossing one of the cut edges could instead reflect in the cut edge to give a path of the same length that doesn't cross the edge.

Since $X$ is at the centre of its face, it makes sense to place the diagram on a grid of equilateral triangles, where the side of the tetrahedron has length 3.



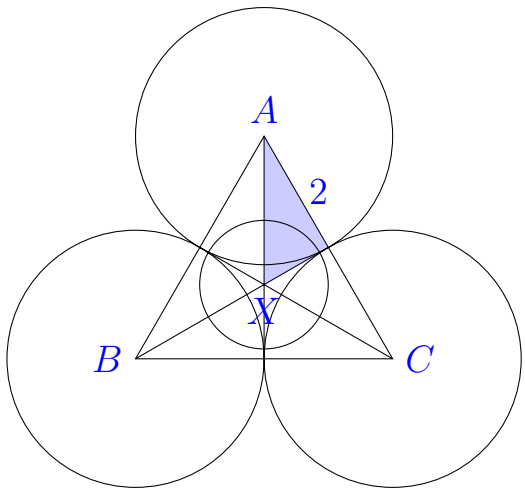

Then as shown, the perpendicular bisectors of each $XV$ (dashed) also lies on this grid. The unshaded area is closer to $X$ than $V$ and the shaded area closer to $V$ than $X$. The shaded area is $\frac{12}{36} = \frac{1}{3}$ of the total surface area,

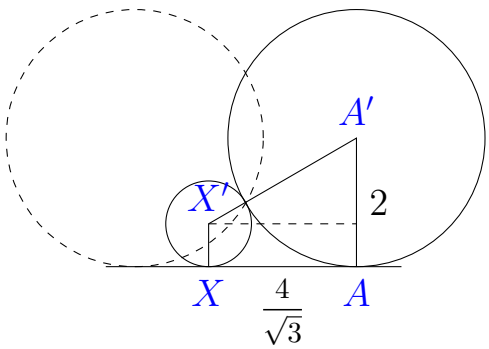
hence (B).

**25.** Three spheres of radius 2 sit on a flat surface touching one another. A smaller sphere sits on the same surface, in the middle and touching all three of the bigger spheres. What is its radius?

(A) $2\sqrt{3}-2\sqrt{2}$ (B) $2\sqrt{\sqrt{3}-\sqrt{2}}$ (C) $\sqrt{3}$ (D) $\frac{3}{2}$ (E) $\frac{2}{3}$

(MW/AK) A top view is shown on the left, with points $A, B, C$ and $X$ being the points where the sphere touches the flat surface. By symmetry, $X$ is at the centre of the equilateral triangle $\triangle ABC$. This creates six triangles with angles $30°$, $60°$ and $90°$ and sides in proportions $1, 2, \sqrt{3}$. Using these ratios in the shaded triangle, $AX = \frac{4}{\sqrt{3}}$.




The diagram on the right show the vertical section through $A$ and $X$, including the centres $A'$ and $X'$ of the two spheres.. In this cross-section, if $x$ is the radius of the smaller circle, then there is a right-angled triangle with sides $2-x$, $\frac{4}{\sqrt{3}}$ and $2+x$ so that

$$(2-x)^2 + \left(\frac{4}{\sqrt{3}}\right)^2 = (2+x)^2$$
$$\frac{16}{3} = (2+x)^2 - (2-x)^2 = 4(2x) = 8x$$
$$x = \frac{2}{3}$$

hence (E).

**26.** How many ways can you select four distinct equally spaced numbers from the set $\{1, ..., 40\}$?

The largest possible difference of a suitable arithmetic progression is $\frac{40-1}{3} = 13$. Therefore, there are $40 - 1 \cdot 3 = 37$ arithmetic progressions with difference 1, $40 - 2 \cdot 3 = 34$ arithmetic progressions with difference 2, $40 - 3 \cdot 3 = 31$ arithmetic progressions with difference 3 and so on up to $40 - 13 \cdot 3 = 1$ arithmetic progression with difference 13. Thus, $1 + 4 + \ldots + 37 = \frac{1}{2} \times 38 \times 13 = 247$,

hence (247).

**27.** Digits $a$, $b$ and $c$ are chosen and then two six-digit numbers are formed, $M$ with digits $abcabc$ and $N$ with digits $ababab$. The ratio $M : N$ is $55 : 54$. What is the 3-digit number $abc$?

---

*Alternative 1*
We get

$$\begin{aligned}
54 \cdot \overline{abcabc} &= 55 \cdot \overline{ababab} \\
54 \cdot \overline{abc}(1000+1) &= 55 \cdot \overline{ab}(10100+1) \\
54 \cdot 1001(10 \cdot \overline{ab} + c) &= 55 \cdot 10101 \cdot \overline{ab} \\
54(10010 \cdot \overline{ab} + 1001c) &= 55 \cdot 10101 \cdot \overline{ab} \\
54054c + 540540 \cdot \overline{ab} &= 555555 \cdot \overline{ab} \\
54054c &= 15015 \cdot \overline{ab}.
\end{aligned}$$

We can reduce the latter by 3003. So we get $18c = 5 \cdot \overline{ab}$. Then, $c = 5$ and $\overline{ab} = 18$, which yields $\overline{abc} = 185$,

hence (185).

*Alternative 2*
Write $M = 1001m$ where $m = abc$ and $N = 10101n$ where $n = ab$. Algebraically, $m = 10n + c$. Then

$$\begin{aligned}
\frac{M}{N} &= \frac{55}{54} \\
54M &= 55N \\
54054m &= 555555n \\
540540n + 54054c &= 555555n \\
54054c &= 15015n \\
1001 \times 3^3 \times 2 \times c &= 1001 \times 5 \times 3 \times n \\
3^2 \times 2 \times c &= 5 \times n
\end{aligned}$$

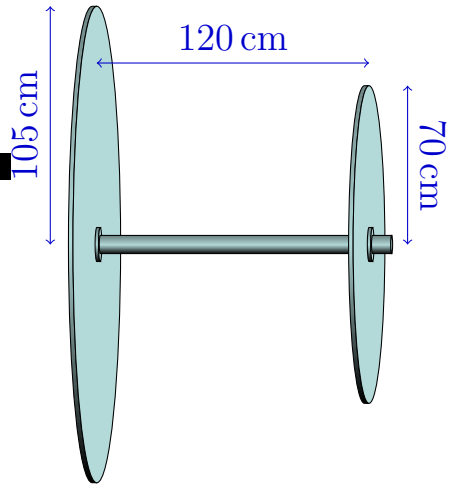
This last number is a common multiple of 18 and 5 so it is a multiple of $\operatorname{lcm}(18, 5) = 90$, say $18c = 5n = 90k$. However, $c < 10$ so $k < 2$ and $k = 1$. Hence $c = 5$, $n = 18$ and the three-digit number $abc$ is 185,

hence (185).

**28.** Two wheels are fixed to an axle as shown.

Due to their different sizes, the two wheels trace two concentric circles when [redacted] rolled on level ground.

In centimetres, what is the radius of the circle traced on the ground by the larger wheel?




---

*Alternative 1*

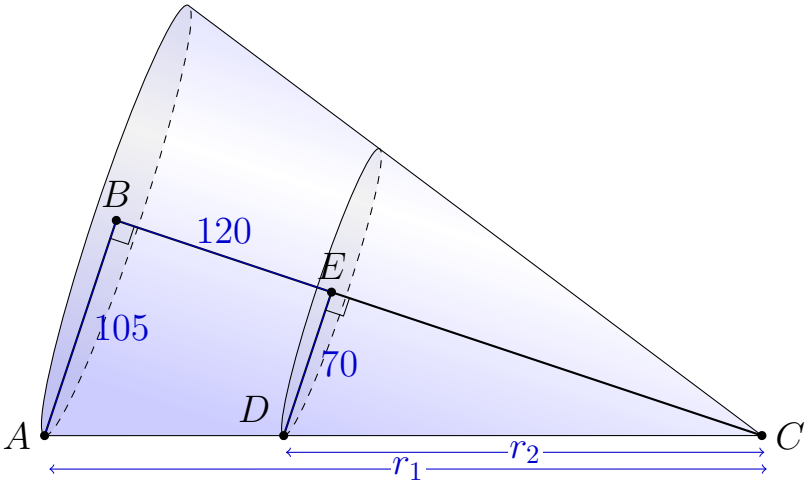
(AK) For any distance rolled, the larger wheel travels $\frac{105}{70} = \frac{3}{2}$ as far as the smaller wheel. So the circumference of the larger traced circle is $\frac{3}{2}$ that of the smaller traced circle, which means the radii are also in the same ratio. Consequently the distance between the two traced circles is half the radius of the smaller circle, and also one-third the radius of the larger circle.

The difference between these two radii is the distance between the two wheel rims, which is $\sqrt{(105-70)^2+120^2} = \sqrt{35^2+120^2} = 5\sqrt{7^2+24^2} =$ [redacted] $5 \times \sqrt{625} = 5 \times 25 = 125\,\text{cm}$. Consequently the larger traced circle has radius $375\,\text{cm}$,

hence (375).

*Alternative 2*

(PC/AK) The pair of wheels can be imagined as part of a cone, and we consider a points $A, B, C, D, E$ in a vertical plane as shown below. The rolling of the wheels is equivalent to rolling the cone with its apex $C$ fixed. Then the radii of the two circles traced by the wheels are $r_1 = AC$ and $r_2 = CD$.




Then $\triangle ABC$ and $\triangle DEC$ are similar with $DE = \frac{2}{3}AB$ so that $EC = \frac{2}{3}BC$. Then $BE = 120 = \frac{1}{3}BC$ so that $BC = 360$ and $EC = 240$.

Rather than using Pythagoras' theorem directly to work out $AC$, observe the ratio $EC : DE = 24 : 7$ in $\triangle DEC$. Then $24^2 + 7^2 = 576 + 49 = 625 = 25^2$ so it follows that both triangles are in the proportions of the Pythagorean triple $(7, 24, 25)$.

With these proportions, $\triangle ABC$ has sides $AB = 15 \times 7 = 105$, $BC = 15 \times 24 = 360$ and $AC = r_1 = 15 \times 25 = 375$,

hence (375).

**29.** Martin the gardener has 3 new vegetable beds, near the kitchen, laundry and shed. Each year he will plant one bed with tomatoes, one with beans, and one with carrots. He needs a schedule for planting that goes for 8 summers.

To balance the disease risk and soil nutrients, his schedule must follow these rules:

- A bed that has tomatoes one summer can't have tomatoes next summer or the one after.
- Carrots can't be in a bed that had beans last summer.

In how many ways can he schedule his vegetable planting for these 8 summers?

---

(AK) In any three-year period, Tomatoes must have one planting in each bed. So they must cycle through the three beds in a three-year cycle, such as shed, laundry, kitchen, shed, laundry, kitchen, etc. There are six ways to have such a cycle, corresponding to the six permutations of {shed, laundry, kitchen}.

Suppose we have such a cycle, and we number the beds 1, 2, 3 depending which year the tomatoes are first planted there. So the plan looks like this:

| Year: | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Bed 1 | T | | | T | | | T | |
| Bed 2 | | T | | | T | | | T |
| Bed 3 | | | T | | | T | | |

We now need to place symbols B and C in each column so that we never see $\boxed{\text{B}}\boxed{\text{C}}$ horizontally. Let $A_n$ be the number of ways to place symbols B and C in columns 1 up to $n$. Clearly $A_1 = 2$ and $A_2 = 3$.

Consider a year $n$, as shown below. Either (i) Beans or (ii) Carrots will follow Tomatoes in the previous year, so we can consider two cases. Once year $n$ is specified, there may be restrictions on year $n-1$ to avoid $\boxed{\text{B}}\boxed{\text{C}}$.

(i)

| $n-2$ | $n-1$ | $n$ |
|---|---|---|
| T | C | C |
| any | T | B |
| any | B | T |

(ii)

| $n-2$ | $n-1$ | $n$ |
|---|---|---|
| T | any | B |
| | T | C |
| | any | T |

As indicated, case (i) can only arise from specific planting schedules up to year $n-1$. However, there is no restriction on year $n-2$, so any plan up to year $n-2$ can be extended to a unique planting in this category. That is, the number of possible schedules in case (i) is $A_{n-2}$.

For case (ii) there is no restriction on which way round the Beans and Carrots are planted in year $n-1$. So the number of ways of scheduling the plantings and ending as in (ii) is the same as the number of ways of scheduling up to year $n-1$, which is $A_{n-1}$.

Consequently $A_n = A_{n-1} + A_{n-2}$, which is the famous Fibonacci recurrence relation. Since we start $A_1 = 2$, the 3rd Fibonacci number, and then $A_3 = 3$, the 4th Fibonacci number, we can deduce that $A_8$ is the 10th Fibonacci number, which is 55.

So for each of the 6 ways of scheduling the tomatoes, there are 55 ways of scheduling the beans and carrots, making a total of $6 \times 55 = 330$ possible ways to schedule the planting,

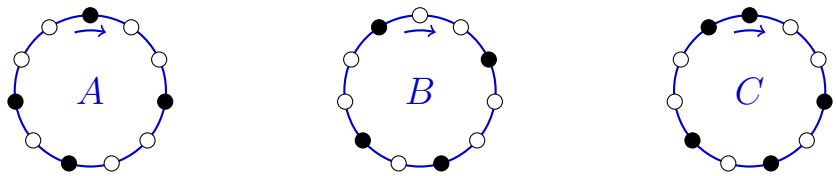
hence (330).

**30.** A percussionist is practising patterns within an 11-beat bar of music. To visualise this, she arranges 11 dots around a circle, with black dots representing a drum hit. She reads this pattern of dots clockwise, starting at the top.

Her patterns have at least one black dot, no two adjacent black dots and two patterns only count as the same if they are the same in every detail, including where the pattern starts in the circle and the direction it is read.

For instance, patterns $A$ and $B$ below are two of her patterns, and they count as different, even though $B$ can be thought of as $A$ starting on a different beat. Pattern $C$ is not one of her patterns, since it has two adjacent black dots.

How many drumming patterns like this are possible?




---

(MW) To compute this, first number the spots $0 \ldots 10$, and then write down the sequences of spots that are allowed.

To keep things within reasonable bounds, count the ones where there is a hit at spot 0 (call these "basic" patterns), and then multiply up accordingly.

For example, for a three-hit pattern we need three numbers 0, $m$ and $n$ such that $2 \leq m$, $m + 2 \leq n$ and $n \leq 9$. When the number of such patterns has been found, multiply by 11 for the various rotations. But now every three-hit pattern has been counted three times (for each of its three hits occurring at spot 0).

This is all made much easier because no allowed pattern has internal symmetries to take account of.

So: patterns of one hit: start with 1 basic pattern, multiply by 11 and divide by 1 to get **11**.

Patterns of two hits. Basic patterns are 02 03 04 *.dots* , 09. There are 8 of them; multiply by 11 and divide by 2 to get **44**.

Patterns of three hits. The basic patterns are

| | | |
|---|---|---|
| 024 | ... | 029 |
| 035 | ... | 039 |
| | ⋮ | |
| 079 | | |

There are $6+5+\ldots 1 = 21$ patterns here. Multiply by 1‘1 and divide by 3 to get **77**.

Patterns of four hits. The basic patterns are

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0246 | ... | 0249 | 0357 | ... | 0359 | 0468 | ... | 0469 | 0579 |
| 0257 | ... | 0259 | 0368 | ... | 0369 | 0479 | | | |
| 0268 | ... | 0269 | 0379 | | | | | | |
| 0279 | | | | | | | | | |

There are $10+6+3+1 = 20$ basic patterns here; multiply by 11 and divide by 4 to get **55**.

Patterns of five hits can be done as above, but the quicker way is to notice that there must be four gaps of 2 and one gap of 3, so the pattern is defined by where the gap of 3 is. There are 11 choices of this, so **11**.

The total number of patterns is $11 + 44 + 77 + 55 + 11 = 198$,

hence (198).

# 4 Answer lists

| MP | UP | J | I | S |
|---|---|---|---|---|
| 1: D | 1: D | 1: A | 1: C | 1: B |
| 2: B | 2: B | 2: B | 2: E | 2: D |
| 3: A | 3: C | 3: A | 3: C | 3: C |
| 4: C | 4: B | 4: A | 4: E | 4: E |
| 5: E | 5: C | 5: B | 5: C | 5: A |
| 6: D | 6: D | 6: C | 6: D | 6: B |
| 7: C | 7: A | 7: E | 7: C | 7: A |
| 8: E | 8: D | 8: A | 8: A | 8: D |
| 9: D | 9: B | 9: D | 9: D | 9: A |
| 10: B | 10: B | 10: D | 10: C | 10: E |
| 11: C | 11: E | 11: C | 11: C | 11: B |
| 12: C | 12: A | 12: C | 12: E | 12: B |
| 13: E | 13: D | 13: D | 13: E | 13: C |
| 14: D | 14: C | 14: D | 14: B | 14: C |
| 15: A | 15: D | 15: B | 15: D | 15: D |
| 16: B | 16: E | 16: B | 16: B | 16: D |
| 17: E | 17: C | 17: C | 17: A | 17: A |
| 18: E | 18: B | 18: E | 18: B | 18: E |
| 19: B | 19: E | 19: A | 19: B | 19: B |
| 20: C | 20: A | 20: B | 20: E | 20: B |
| 21: D | 21: E | 21: D | 21: E | 21: D |
| 22: E | 22: D | 22: A | 22: D | 22: A |
| 23: A | 23: C | 23: B | 23: A | 23: C |
| 24: B | 24: D | 24: B | 24: C | 24: B |
| 25: C | 25: D | 25: E | 25: C | 25: E |
| 26: 214 | 26: 285 | 26: 541 | 26: 301 | 26: 247 |
| 27: 285 | 27: 252 | 27: 34 | 27: 97 | 27: 185 |
| 28: 23 | 28: 711 | 28: 351 | 28: 62 | 28: 375 |
| 29: 674 | 29: 630 | 29: 164 | 29: 375 | 29: 330 |
| 30: 630 | 30: 925 | 30: 810 | 30: 170 | 30: 198 |

| | | | | |
|---|---|---|---|---|
| A: 3 | A: 3 | A: 6 | A: 3 | A: 5 |
| B: 5 | B: 5 | B: 7 | B: 4 | B: 7 |
| C: 6 | C: 5 | C: 4 | C: 8 | C: 4 |
| D: 5 | D: 8 | D: 5 | D: 4 | D: 5 |
| E: 6 | E: 4 | E: 3 | E: 6 | E: 4 |